# model Decision


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## Imports

In [183…

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.neural_network import MLPClassifier

import pandas as pd
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier as KNN
from sklearn.discriminant_analysis import QuadraticDiscriminantAnalysis as QDA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis as LDA
from sklearn.naive_bayes import GaussianNB as GNB
from sklearn.tree import DecisionTreeClassifier as DT
from sklearn import tree
from sklearn.ensemble import RandomForestClassifier as RFC
from sklearn.linear_model import LogisticRegression
from sklearn.svm import SVC
import keras
from keras import layers
from xgboost import XGBClassifier
from sklearn.metrics import accuracy_score, ConfusionMatrixDisplay, confusion_ma
from sklearn.cluster import KMeans
from sklearn.mixture import GaussianMixture
from sklearn.decomposition import PCA
import warnings
from sklearn.preprocessing import label_binarize
from sklearn.discriminant_analysis import StandardScaler
```

---

## helpfull code

In [184…

```
def evaluate_model_metrics(model, model_name,
                           x_train, y_train,
                           x_test, y_test):

    y_pred_train = model.predict(x_train)
    y_pred_test  = model.predict(x_test)

    report_dict = classification_report(
        y_test,
        y_pred_test,
        output_dict=True
    )

    report_df = pd.DataFrame(report_dict).T
```

```
    report_df = report_df.loc[
        [str(c) for c in sorted(y_test.unique())] +
        ["accuracy"]
    ]


    display(
        report_df.style
            .format("{:.2%}")
            .set_caption(f"{model_name} Classification Report")
    )
```

In [185…

```
def plot_confusion_matrices(model, model_name,
                            x_train, y_train,
                            x_test, y_test):

    y_pred_train = model.predict(x_train)
    y_pred_test  = model.predict(x_test)

    cm_train = confusion_matrix(y_train, y_pred_train)
    cm_test  = confusion_matrix(y_test, y_pred_test)

    fig, axes = plt.subplots(1, 2, figsize=(12,5))

    ConfusionMatrixDisplay(cm_train).plot(ax=axes[0], colorbar=False)
    axes[0].set_title(f"{model_name} - Train")

    ConfusionMatrixDisplay(cm_test).plot(ax=axes[1], colorbar=False)
    axes[1].set_title(f"{model_name} - Test")

    plt.tight_layout()
    plt.show()
```

In [ ]:

```
def plot_roc_curve(model, model_name, x_test, y_test):

    if hasattr(model, "predict_proba"):
        y_score = model.predict_proba(x_test)[:, 1]
    elif hasattr(model, "decision_function"):
        y_score = model.decision_function(x_test)
    else:
        raise ValueError(f"{model_name} does not support predict_proba or decisi

    fpr, tpr, _ = roc_curve(y_test, y_score)
    roc_auc = auc(fpr, tpr)

    plt.figure(figsize=(7,5))
    plt.plot(fpr, tpr, label=f"{model_name} (AUC = {roc_auc:.3f})")
    plt.plot([0, 1], [0, 1], "--")  # baseline random
    plt.xlabel("False Positive Rate")
    plt.ylabel("True Positive Rate")
    plt.title(f"ROC Curve - {model_name}")
    plt.grid(alpha=0.3)
    plt.legend()
    plt.tight_layout()
    plt.show()
```

In [286…

```
def plot_feature_transport_pies(df, feature_col, target_col="Transported"):
```

```
    fig, axes = plt.subplots(1, 3, figsize=(16,5))

    # ---------- Pie 1 ----------
    data1 = df[df[feature_col] == 1][target_col].value_counts().sort_index()
    axes[0].pie(
        data1,
        labels=["Not Transported", "Transported"],
        autopct="%1.1f%%",
        startangle=90
    )
    axes[0].set_title(f"{feature_col}=1 → Transported")

    # ---------- Pie 2 ----------
    data2 = df[df[feature_col] == 0][target_col].value_counts().sort_index()
    axes[1].pie(
        data2,
        labels=["Not Transported", "Transported"],
        autopct="%1.1f%%",
        startangle=90
    )
    axes[1].set_title(f"{feature_col}=0 → Transported")

    # ---------- Pie 3 ----------
    data3 = df[df[target_col] == 1][feature_col].value_counts().sort_index()
    axes[2].pie(
        data3,
        labels=[f"{feature_col}=0", f"{feature_col}=1"],
        autopct="%1.1f%%",
        startangle=90
    )
    axes[2].set_title(f"Among Transported → {feature_col}")

    plt.tight_layout()
    plt.show()
```

---

# Load Data

In [187…

```
spaceship_data = pd.read_csv('final_data.csv')
```

# split to x and y

In [282…

```
y = spaceship_data["Transported"].astype(int)
x = spaceship_data.drop(columns=["Transported"])
x_copy = x.copy()
```

# Train-Test Split

In [189…

```
x_train, x_test, y_train, y_test = train_test_split(
    x, y, test_size=0.2, random_state=42
)
```

# Scaling

In [190…
```
scaler = StandardScaler()
x_train = scaler.fit_transform(x_train)
x_test  = scaler.transform(x_test)
```

# הרצת מודלים שנבחנו:

## שמירת תוצאות

In [191…
```
results = []
```

## KNN

### בנייה

In [192…
```
knn = KNN(
    n_neighbors=43,
    metric="euclidean",
    weights="uniform"
)

knn.fit(x_train, y_train)
```

Out[192…
```
KNeighborsClassifier
KNeighborsClassifier(metric='euclidean', n_neighbors=43)
```

### דיוק אימון ומבחן

In [193…
```
y_pred_train_knn = knn.predict(x_train)
train_acc_knn = accuracy_score(y_train, y_pred_train_knn)

y_pred_test_knn = knn.predict(x_test)
test_acc_knn = accuracy_score(y_test, y_pred_test_knn)
```

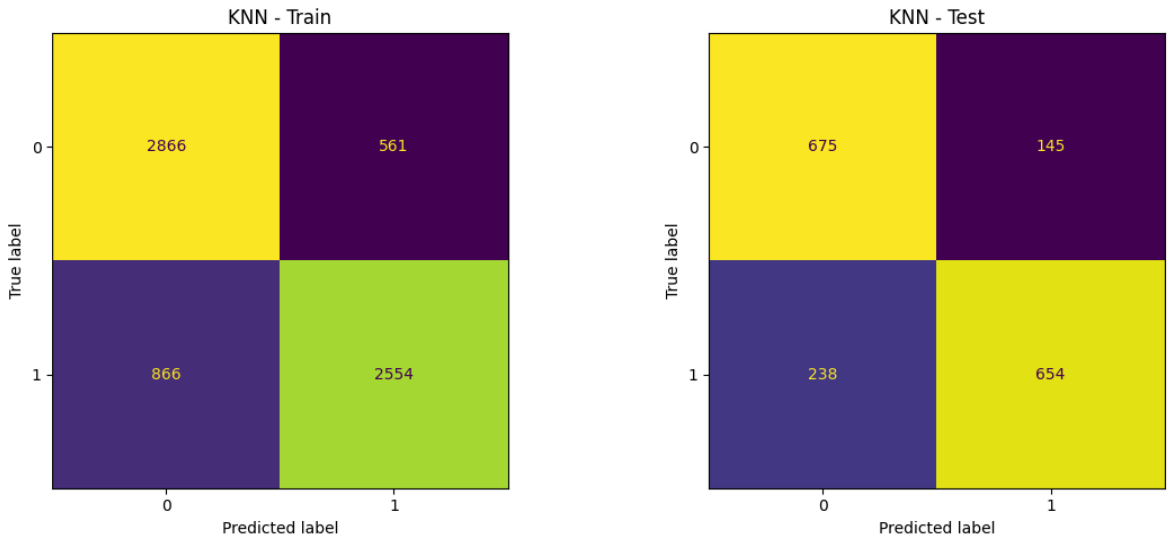
In [194…
```
print("="*40)
print("BEST KNN – TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_knn*100:.2f}%")
print("="*40)

print("="*40)
print("BEST KNN – TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_knn*100:.2f}%")
print("="*40)
```

```
========================================
BEST KNN — TRAIN SET PERFORMANCE
========================================
Accuracy : 79.16%
========================================
========================================
BEST KNN — TEST SET PERFORMANCE
========================================
Accuracy : 77.63%
========================================
```

## Confusion matrix

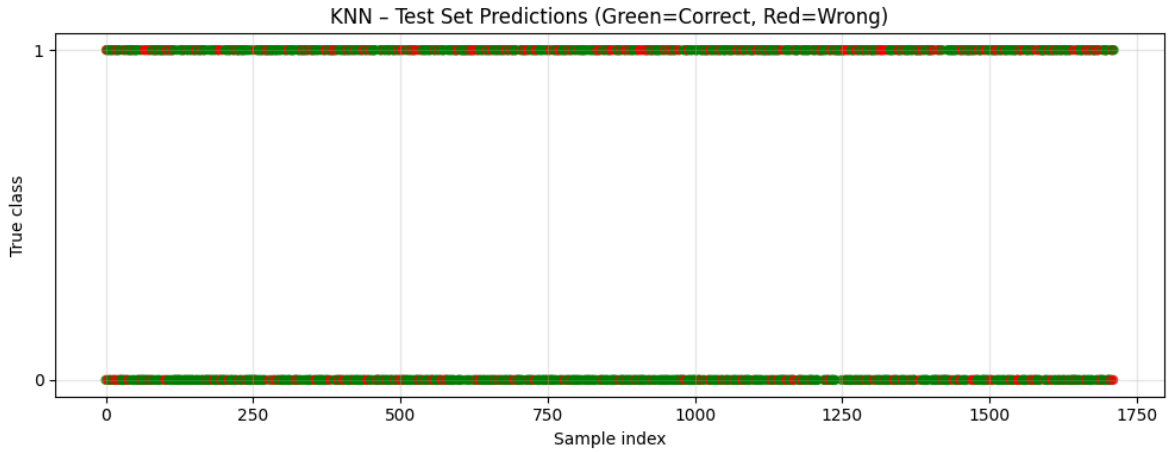
In [195…

```
plot_confusion_matrices(knn, "KNN", x_train, y_train, x_test, y_test)
```




In [196…

```
correct = (y_pred_test_knn == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("KNN – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"KNN Test Accuracy: {100*test_acc_knn:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
KNN Test Accuracy: 77.63%
Errors: 383 / 1712
```

### accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [197…

```
evaluate_model_metrics(knn, "KNN", x_train, y_train, x_test, y_test)
```

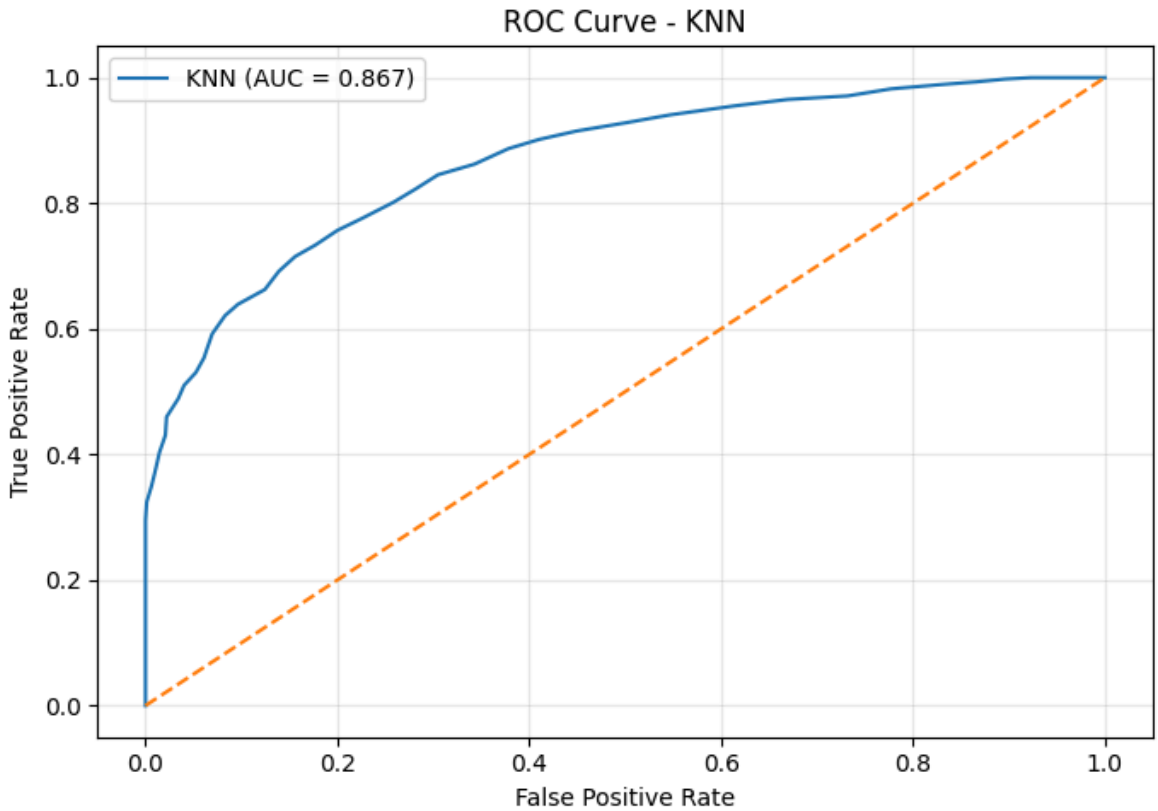
KNN Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 73.93% | 82.32% | 77.90% | 82000.00% |
| **1** | 81.85% | 73.32% | 77.35% | 89200.00% |
| **accuracy** | 77.63% | 77.63% | 77.63% | 77.63% |

### roc curve

In [198…

```
auc_val = plot_roc_curve(knn, "KNN", x_test, y_test)
```



שמירה

In [199…
```
results.append({
    "Model": "KNN",
    "Train Accuracy": train_acc_knn,
    "Test Accuracy": test_acc_knn
})
```

---

# QDA

## בנייה

In [200…
```
qda = QDA()
qda.fit(x_train, y_train)
```

```
c:\Users\Liad M\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn
\discriminant_analysis.py:1024: LinAlgWarning: The covariance matrix of class 0 i
s not full rank. Increasing the value of parameter `reg_param` might help reducin
g the collinearity.
  warnings.warn(
c:\Users\Liad M\AppData\Local\Programs\Python\Python312\Lib\site-packages\sklearn
\discriminant_analysis.py:1024: LinAlgWarning: The covariance matrix of class 1 i
s not full rank. Increasing the value of parameter `reg_param` might help reducin
g the collinearity.
  warnings.warn(
```

Out[200…
```
▾ QuadraticDiscriminantAnalysis
QuadraticDiscriminantAnalysis()
```

## דיוק אימון ומבחן

In [201…
```
y_pred_train_qda = qda.predict(x_train)
train_acc_qda = accuracy_score(y_train, y_pred_train_qda)

y_pred_test_qda = qda.predict(x_test)
test_acc_qda = accuracy_score(y_test, y_pred_test_qda)
```

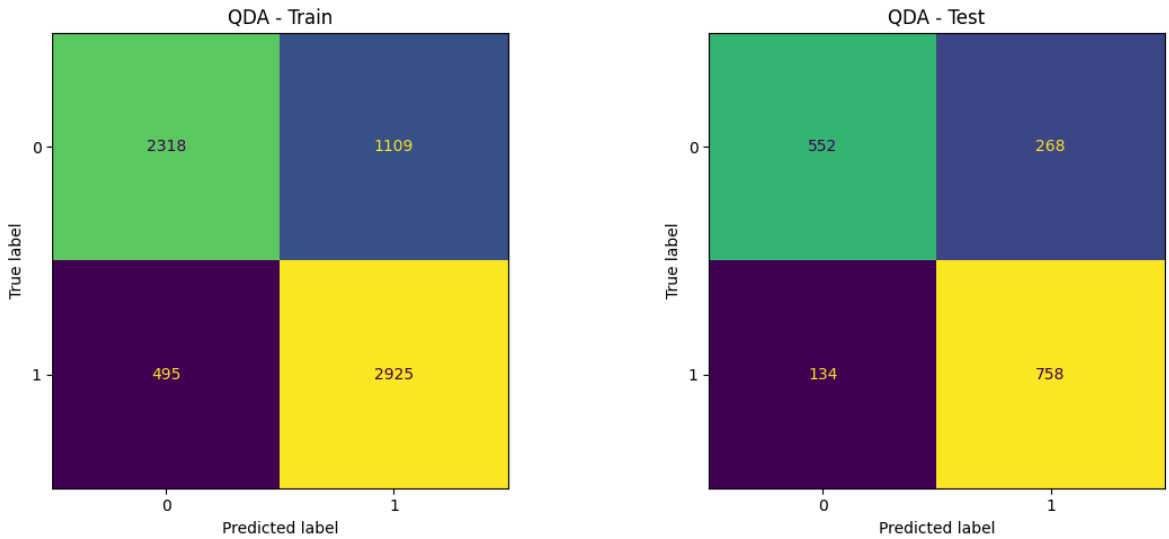
In [202…
```
print("="*40)
print("BEST QDA — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_qda*100:.2f}%")
print("="*40)

print("="*40)
print("BEST QDA — TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_qda*100:.2f}%")
print("="*40)
```

```
========================================
BEST QDA — TRAIN SET PERFORMANCE
========================================
Accuracy : 76.57%
========================================
========================================
BEST QDA — TEST SET PERFORMANCE
========================================
Accuracy : 76.52%
========================================
```

## Confusion matrix

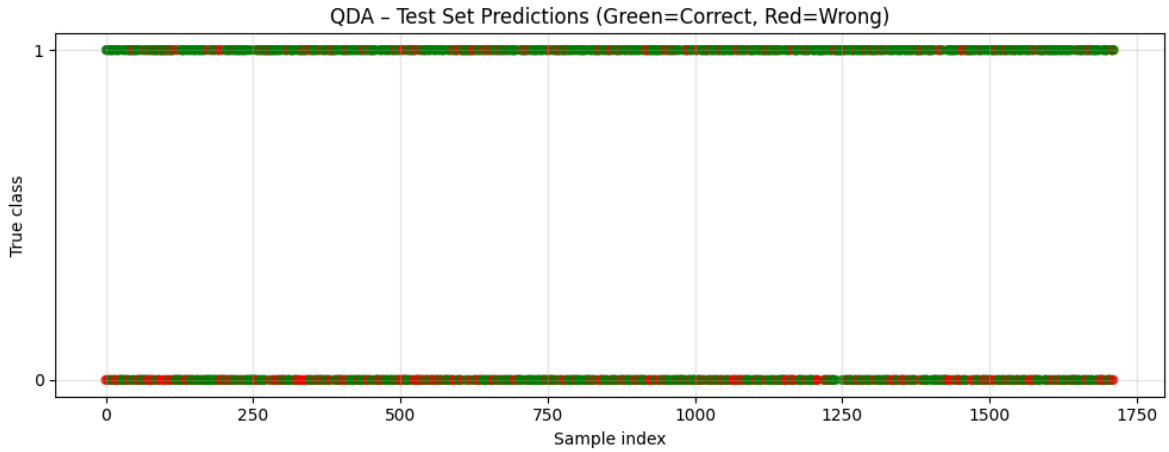
In [203…

```
plot_confusion_matrices(qda, "QDA", x_train, y_train, x_test, y_test)
```




In [204…

```
correct = (y_pred_test_qda == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("QDA – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"QDA Test Accuracy: {100*test_acc_qda:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
QDA Test Accuracy: 76.52%
Errors: 402 / 1712
```

### accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [205…

```
evaluate_model_metrics(qda, "QDA", x_train, y_train, x_test, y_test)
```

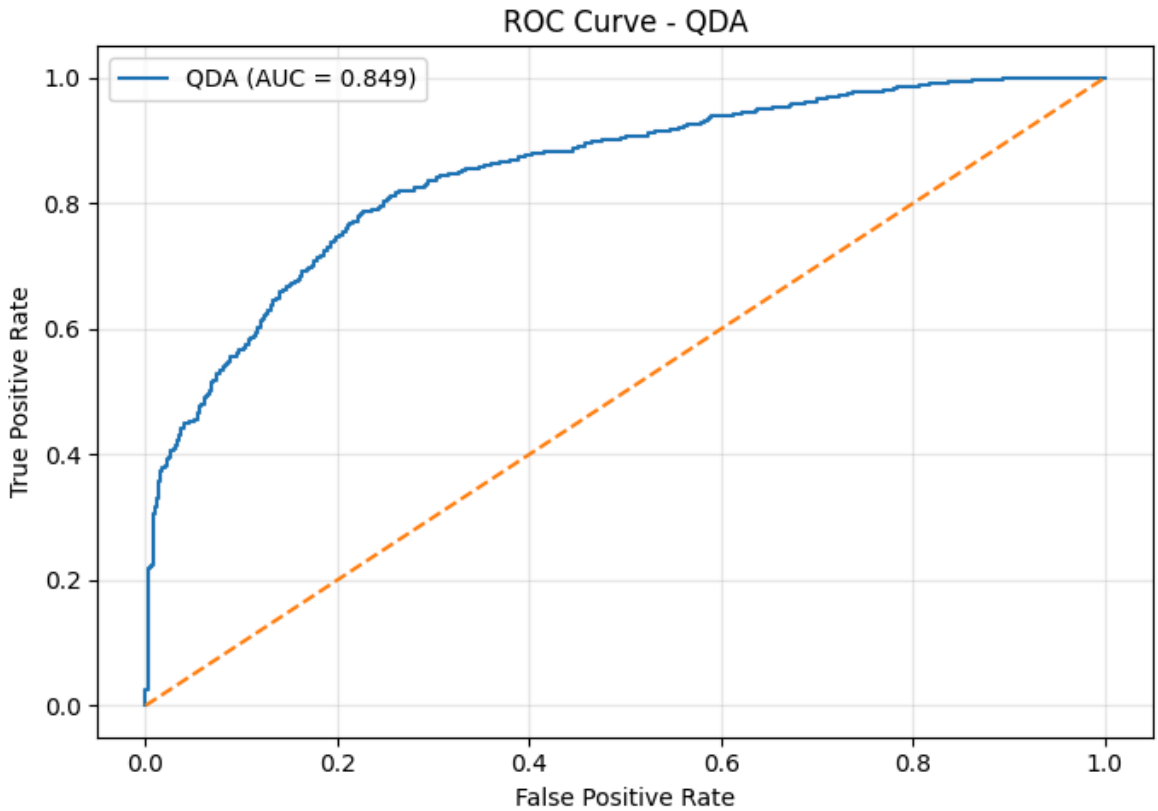
QDA Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 80.47% | 67.32% | 73.31% | 82000.00% |
| **1** | 73.88% | 84.98% | 79.04% | 89200.00% |
| **accuracy** | 76.52% | 76.52% | 76.52% | 76.52% |

### roc curve

In [206…

```
auc_val = plot_roc_curve(qda, "QDA", x_test, y_test)
```



שמירה

In [207…
```
results.append({
    "Model": "QDA",
    "Train Accuracy": train_acc_qda,
    "Test Accuracy": test_acc_qda
})
```

---

# LDA

## בנייה

In [208…
```
lda = LDA()
lda.fit(x_train, y_train)
```

Out[208…
```
▾ LinearDiscriminantAnalysis
LinearDiscriminantAnalysis()
```

## דיוק אימון ומבחן

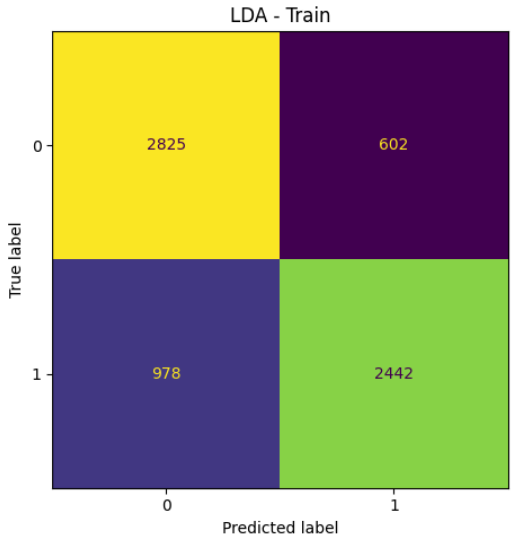
In [209…
```
y_pred_train_lda = lda.predict(x_train)
train_acc_lda = accuracy_score(y_train, y_pred_train_lda)

y_pred_test_lda = lda.predict(x_test)
test_acc_lda = accuracy_score(y_test, y_pred_test_lda)
```

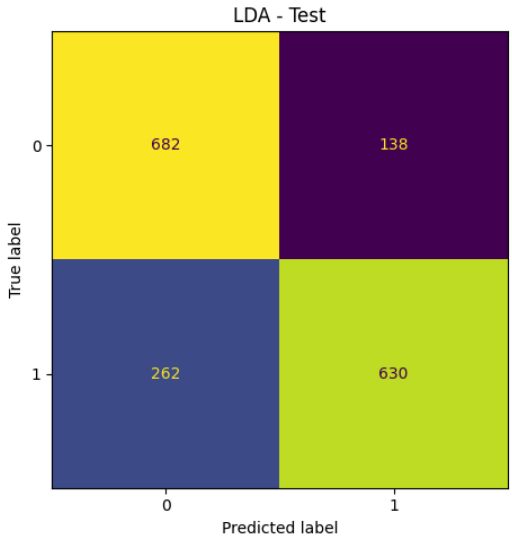
In [210…
```
print("="*40)
print("BEST LDA – TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_lda*100:.2f}%")
print("="*40)

print("="*40)
print("BEST LDA – TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_lda*100:.2f}%")
print("="*40)
```

```
========================================
BEST LDA – TRAIN SET PERFORMANCE
========================================
Accuracy : 76.92%
========================================
========================================
BEST LDA – TEST SET PERFORMANCE
========================================
Accuracy : 76.64%
========================================
```

## Confusion matrix

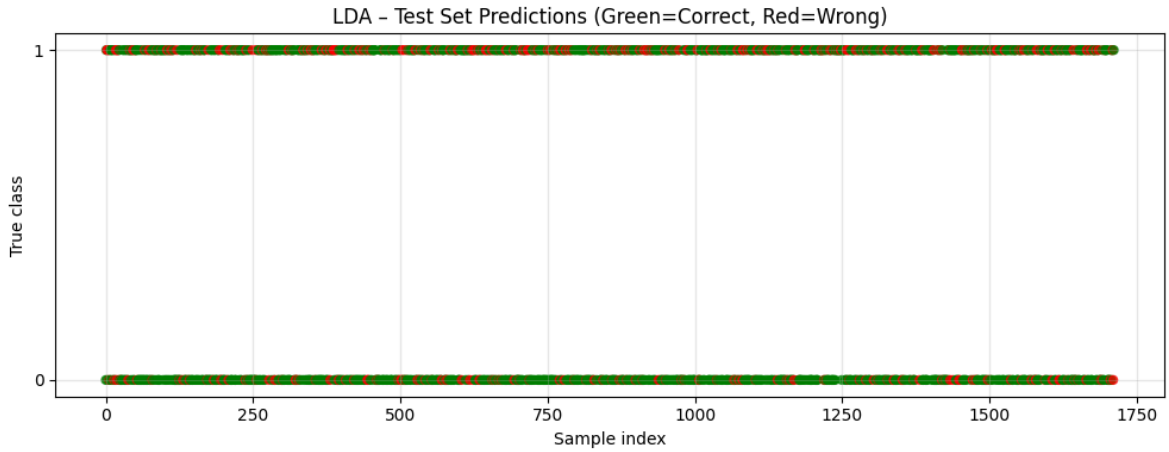
In [211…
```
plot_confusion_matrices(lda, "LDA", x_train, y_train, x_test, y_test)
```




In [212…

```
correct = (y_pred_test_lda == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("LDA – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"LDA Test Accuracy: {100*test_acc_lda:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
LDA Test Accuracy: 76.64%
Errors: 400 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [213…

```
evaluate_model_metrics(lda, "LDA", x_train, y_train, x_test, y_test)
```

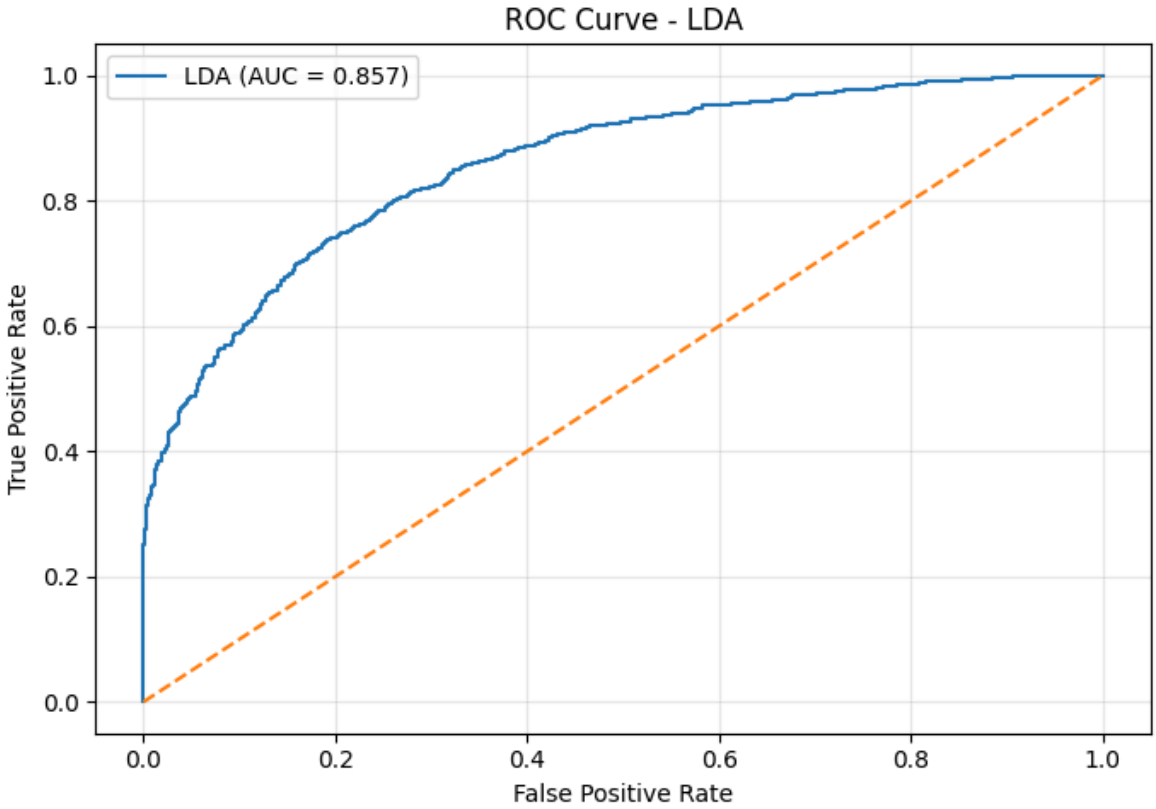
LDA Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 72.25% | 83.17% | 77.32% | 82000.00% |
| **1** | 82.03% | 70.63% | 75.90% | 89200.00% |
| **accuracy** | 76.64% | 76.64% | 76.64% | 76.64% |

### roc curve

In [214…

```
auc_val = plot_roc_curve(lda, "LDA", x_test, y_test)
```




שמירה

In [215…

```
results.append({
    "Model": "LDA",
    "Train Accuracy": train_acc_lda,
    "Test Accuracy": test_acc_lda
})
```

---

## GNB

### בנייה

In [216…

```
gnb = GNB()
gnb.fit(x_train, y_train)
```

Out[216…

```
GaussianNB
GaussianNB()
```

## דיוק אימון ומבחן

In [217…

```
y_pred_train_gnb = gnb.predict(x_train)
train_acc_gnb = accuracy_score(y_train, y_pred_train_gnb)

y_pred_test_gnb = gnb.predict(x_test)
test_acc_gnb = accuracy_score(y_test, y_pred_test_gnb)
```

In [218…

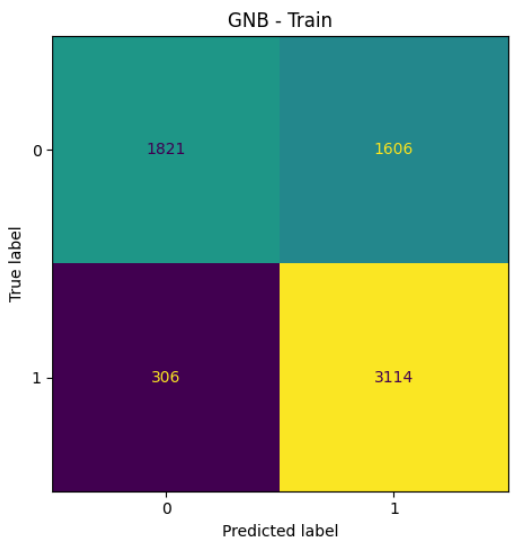
```
print("="*40)
print("BEST GNB – TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_gnb*100:.2f}%")
print("="*40)

print("="*40)
print("BEST GNB – TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_gnb*100:.2f}%")
print("="*40)
```

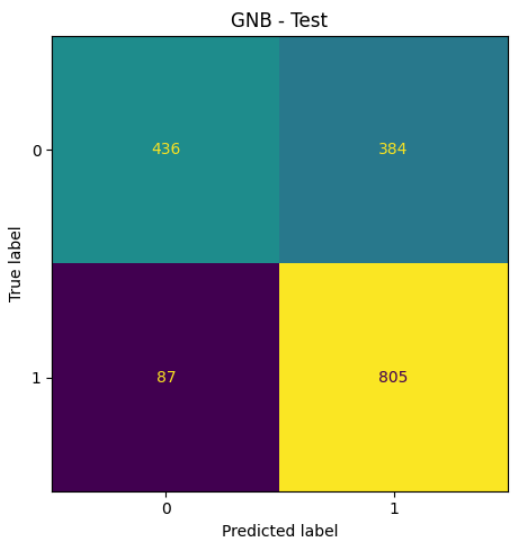
```
========================================
BEST GNB – TRAIN SET PERFORMANCE
========================================
Accuracy : 72.08%
========================================
========================================
BEST GNB – TEST SET PERFORMANCE
========================================
Accuracy : 72.49%
========================================
```

## Confusion matrix

In [219…

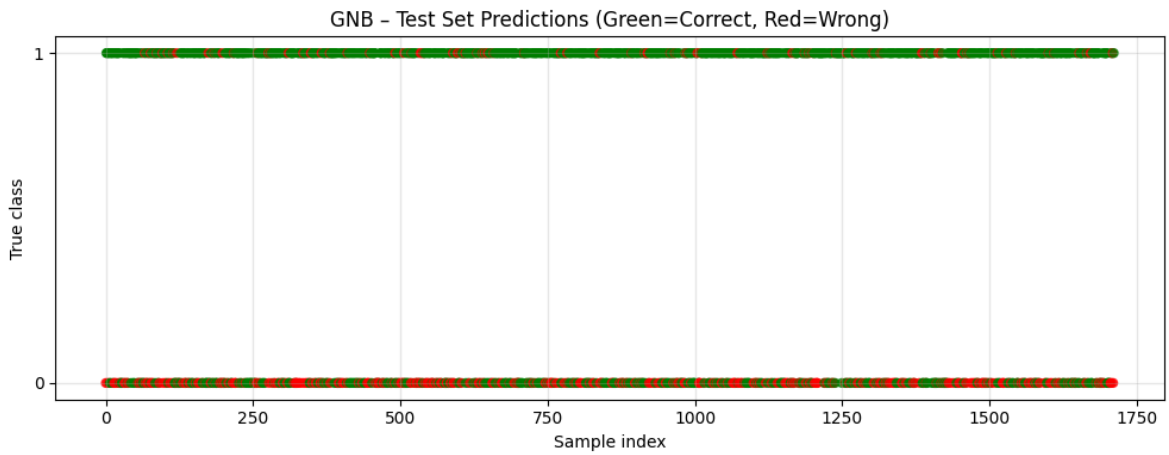
```
plot_confusion_matrices(gnb, "GNB", x_train, y_train, x_test, y_test)
```



In [220…

```
correct = (y_pred_test_gnb == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("GNB – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"GNB Test Accuracy: {100*test_acc_gnb:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
GNB Test Accuracy: 72.49%
Errors: 471 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [221…

```
evaluate_model_metrics(gnb, "GNB", x_train, y_train, x_test, y_test)
```

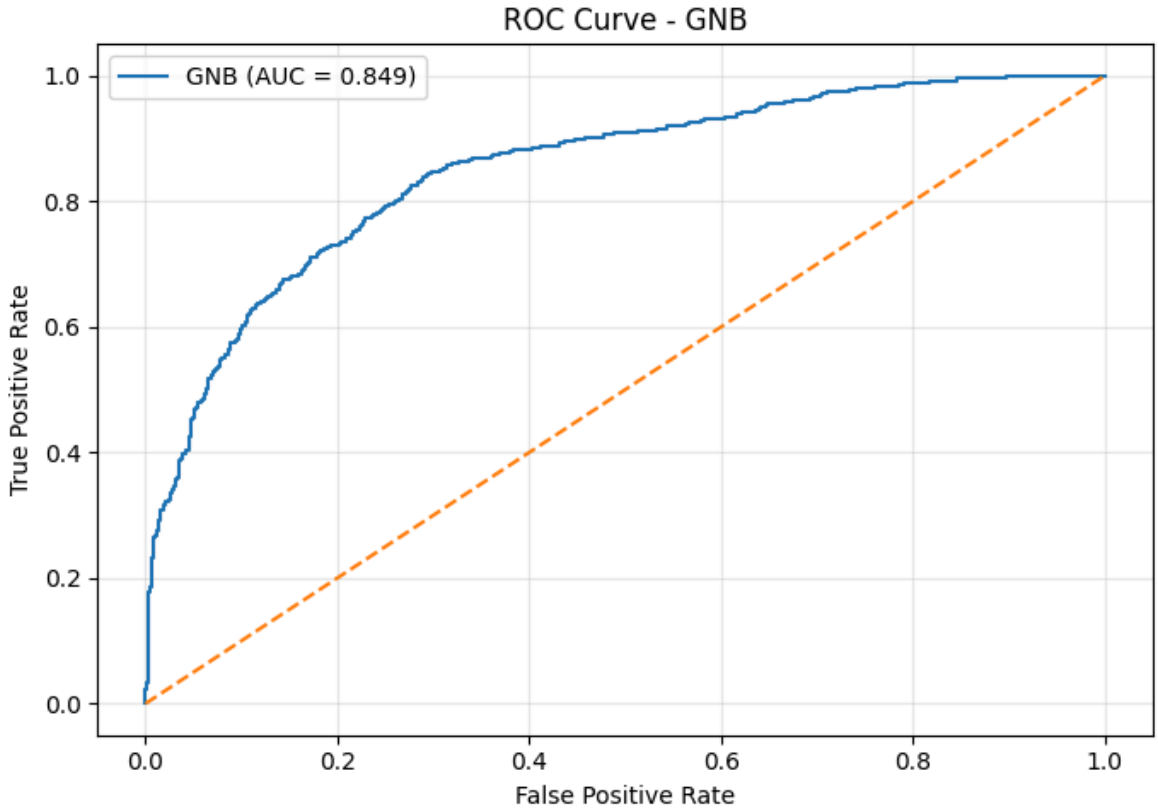
GNB Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 83.37% | 53.17% | 64.93% | 82000.00% |
| **1** | 67.70% | 90.25% | 77.37% | 89200.00% |
| **accuracy** | 72.49% | 72.49% | 72.49% | 72.49% |

## roc curve

In [222…

```
auc_val = plot_roc_curve(gnb, "GNB", x_test, y_test)
```




שמירה

In [223…

```
results.append({
    "Model": "GNB",
    "Train Accuracy": train_acc_gnb,
    "Test Accuracy": test_acc_gnb
})
```

---

## Decision Tree

### בנייה

In [224…

```
dt = DT(
    criterion='gini',
    max_depth=7,
    min_samples_leaf=87,
    min_samples_split=88,
    max_leaf_nodes=38
)

dt.fit(x_train, y_train)
```

Out[224…

```
DecisionTreeClassifier
DecisionTreeClassifier(max_depth=7, max_leaf_nodes=38, min_samples_leaf
=87,
                       min_samples_split=88)
```

## דיוק אימון ומבחן

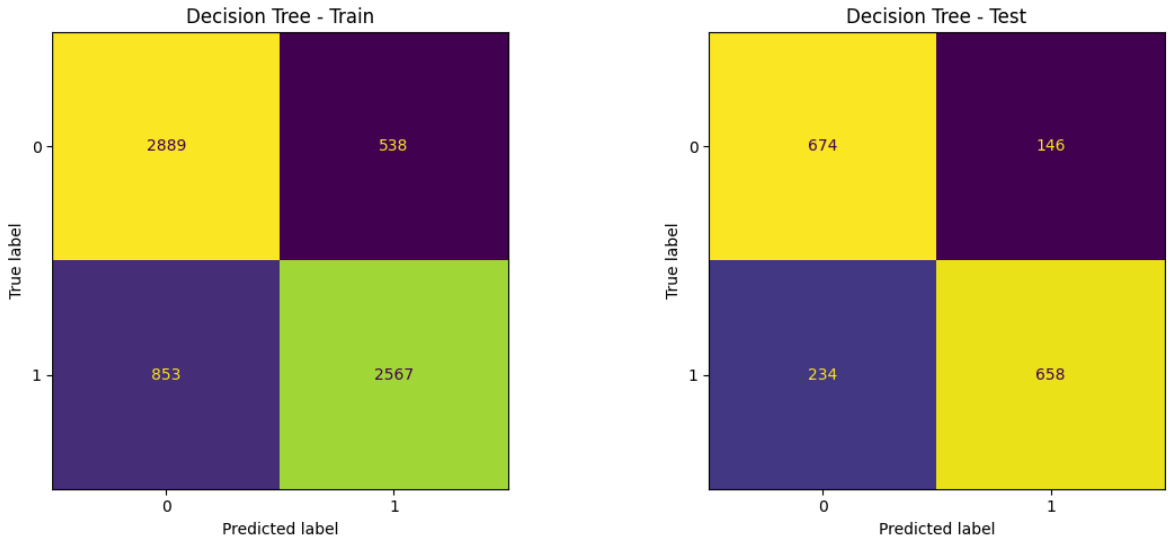
In [225…
```
y_pred_train_dt = dt.predict(x_train)
train_acc_dt = accuracy_score(y_train, y_pred_train_dt)

y_pred_test_dt = dt.predict(x_test)
test_acc_dt = accuracy_score(y_test, y_pred_test_dt)
```

In [226…
```
print("="*40)
print("BEST DT – TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_dt*100:.2f}%")
print("="*40)

print("="*40)
print("BEST DT – TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_dt*100:.2f}%")
print("="*40)
```

```
========================================
BEST DT – TRAIN SET PERFORMANCE
========================================
Accuracy : 79.68%
========================================
========================================
BEST DT – TEST SET PERFORMANCE
========================================
Accuracy : 77.80%
========================================
```

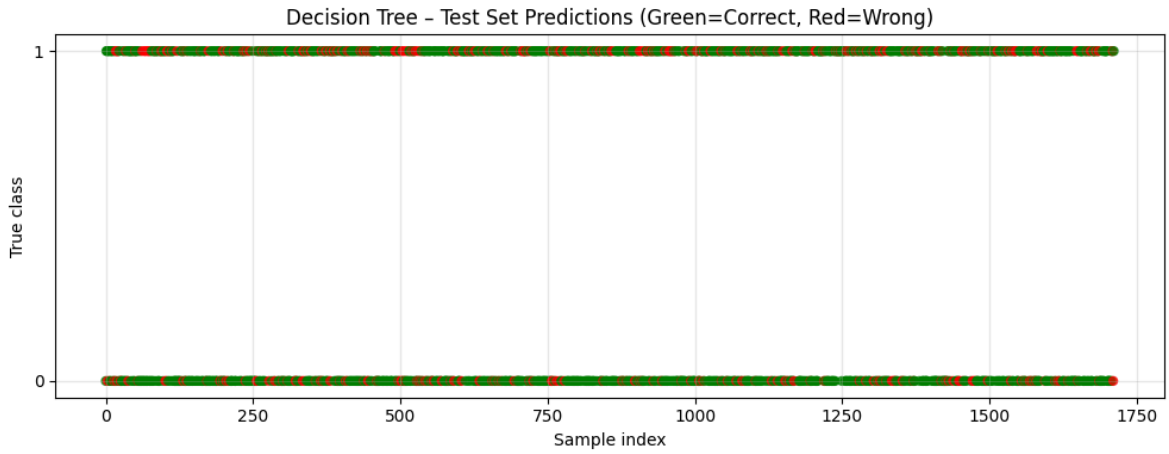
## Confusion matrix

In [227…
```
plot_confusion_matrices(dt, "Decision Tree", x_train, y_train, x_test, y_test)
```




In [228…
```
correct = (y_pred_test_dt == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
```

```
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("Decision Tree – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Decision Tree Test Accuracy: {100*test_acc_dt:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
Decision Tree Test Accuracy: 77.80%
Errors: 380 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [229…

```
evaluate_model_metrics(dt, "Decision Tree", x_train, y_train, x_test, y_test)
```

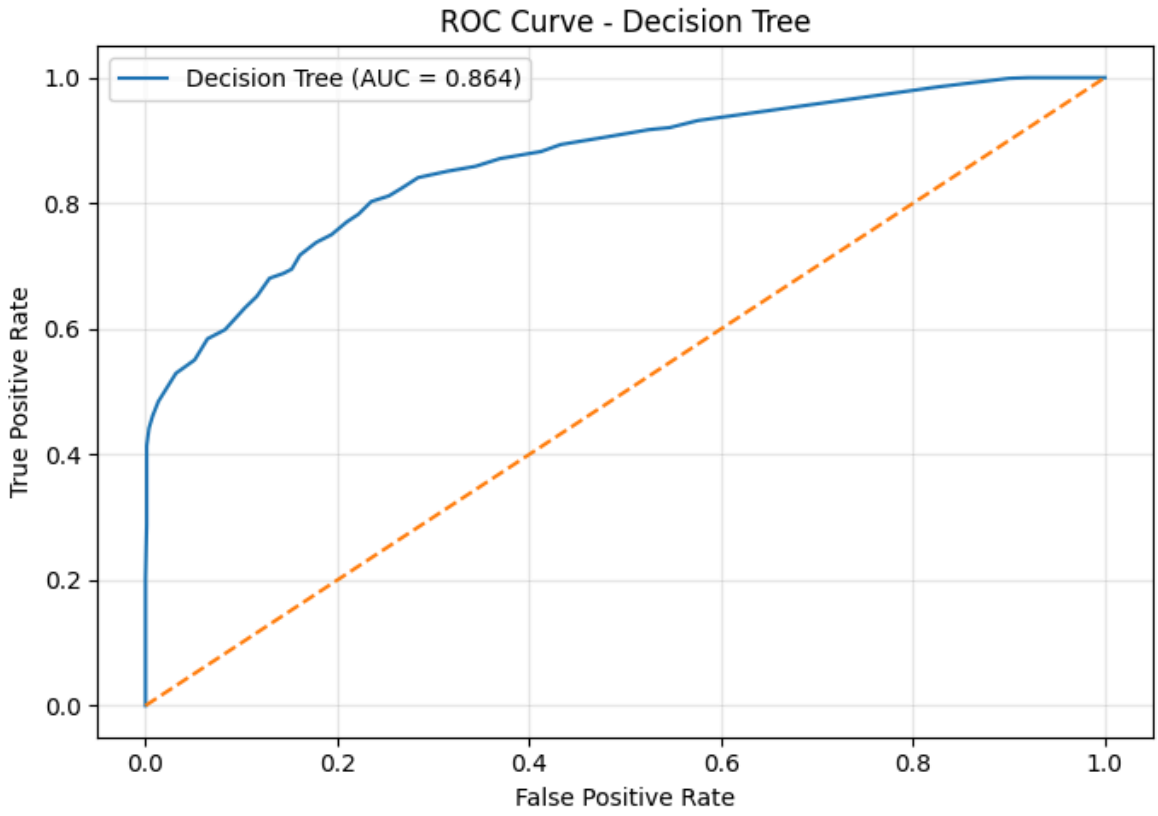
Decision Tree Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 74.23% | 82.20% | 78.01% | 82000.00% |
| **1** | 81.84% | 73.77% | 77.59% | 89200.00% |
| **accuracy** | 77.80% | 77.80% | 77.80% | 77.80% |

## roc curve

In [230…

```
auc_val = plot_roc_curve(dt, "Decision Tree", x_test, y_test)
```




שמירה

In [231…

```
results.append({
    "Model": "Decision Tree",
    "Train Accuracy": train_acc_dt,
    "Test Accuracy": test_acc_dt
})
```

---

## Random Forest

### בנייה

In [232…

```
rf = RFC(
    n_estimators=250,
    random_state=42,
    max_depth=21,
    max_features='sqrt',
    min_samples_leaf=7
)
rf.fit(x_train, y_train)
```

Out[232…

```
RandomForestClassifier
RandomForestClassifier(max_depth=21, min_samples_leaf=7, n_estimators=250,
                       random_state=42)
```

## דיוק אימון ומבחן

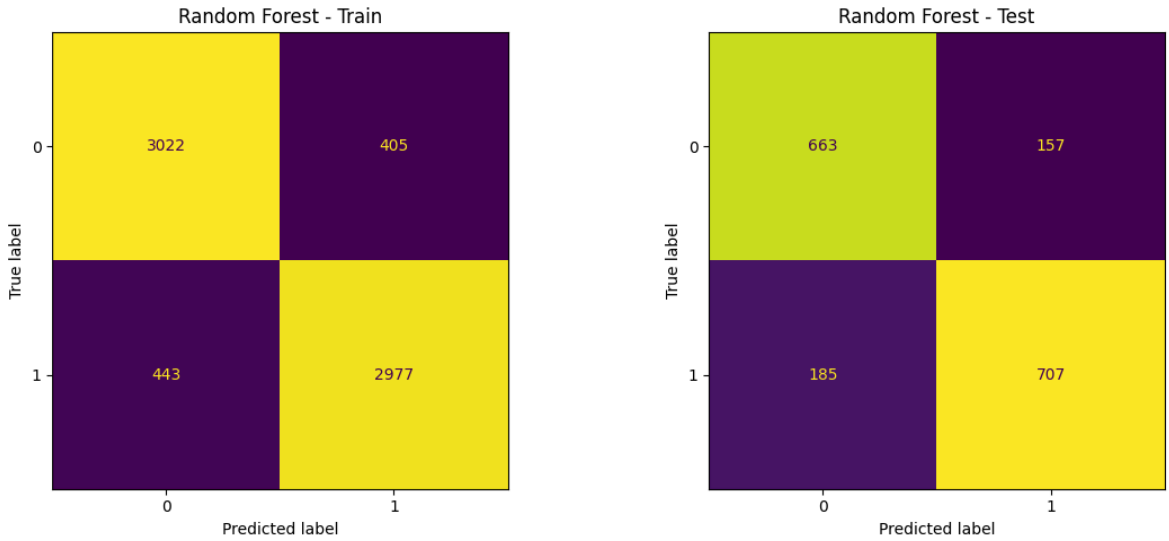
In [233…
```
y_pred_train_rf = rf.predict(x_train)
train_acc_rf = accuracy_score(y_train, y_pred_train_rf)

y_pred_test_rf = rf.predict(x_test)
test_acc_rf = accuracy_score(y_test, y_pred_test_rf)
```

In [234…
```
print("="*40)
print("BEST RF – TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_rf*100:.2f}%")
print("="*40)

print("="*40)
print("BEST RF – TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_rf*100:.2f}%")
print("="*40)
```

```
========================================
BEST RF – TRAIN SET PERFORMANCE
========================================
Accuracy : 87.62%
========================================
========================================
BEST RF – TEST SET PERFORMANCE
========================================
Accuracy : 80.02%
========================================
```

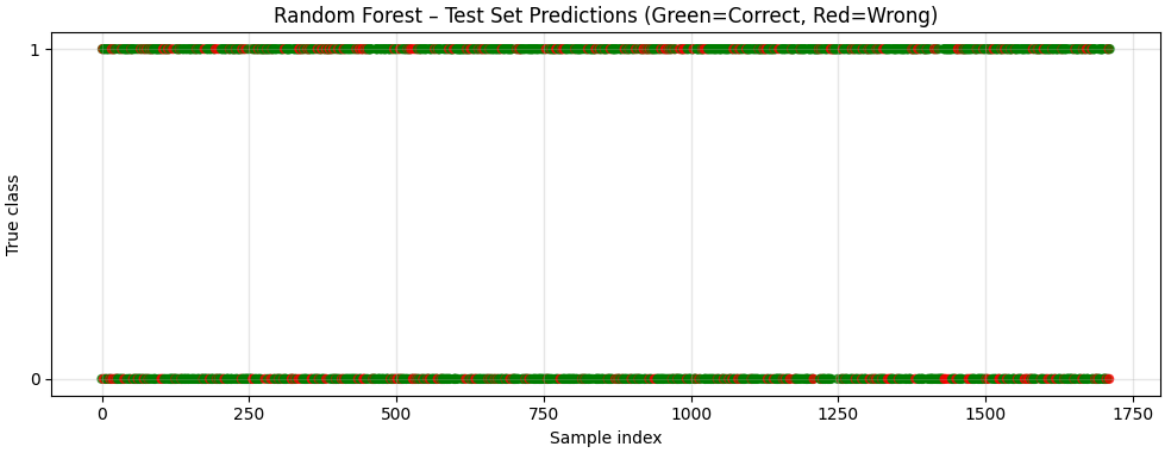
## Confusion matrix

In [235…
```
plot_confusion_matrices(rf, "Random Forest", x_train, y_train, x_test, y_test)
```




In [236…
```
correct = (y_pred_test_rf == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
```

```
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("Random Forest – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Random Forest Test Accuracy: {100*test_acc_rf:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
Random Forest Test Accuracy: 80.02%
Errors: 342 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [237…

```
evaluate_model_metrics(rf, "Random Forest", x_train, y_train, x_test, y_test)
```

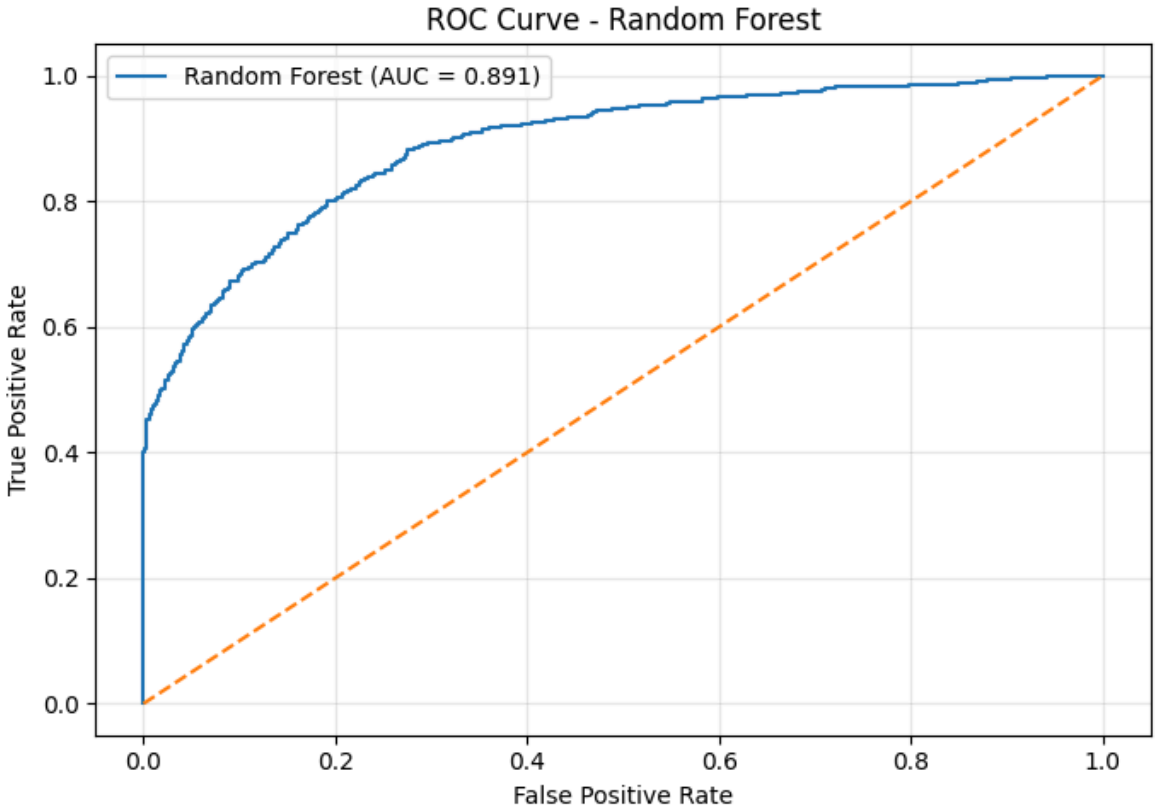
Random Forest Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 78.18% | 80.85% | 79.50% | 82000.00% |
| **1** | 81.83% | 79.26% | 80.52% | 89200.00% |
| **accuracy** | 80.02% | 80.02% | 80.02% | 80.02% |

## roc curve

In [238…

```
auc_val = plot_roc_curve(rf, "Random Forest", x_test, y_test)
```




שמירה

In [239…

```
results.append({
    "Model": "Random Forest",
    "Train Accuracy": train_acc_rf,
    "Test Accuracy": test_acc_rf
})
```

---

# SVM

## בנייה

In [240…

```
svm = SVC(
    kernel='poly',
    C=1000,
    degree=2,
    random_state=42
)
svm.fit(x_train, y_train)
```

Out[240…

```
SVC
SVC(C=1000, degree=2, kernel='poly', random_state=42)
```

## דיוק אימון ומבחן

In [241…

```
y_pred_train_svm = svm.predict(x_train)
train_acc_svm = accuracy_score(y_train, y_pred_train_svm)
```

```
y_pred_test_svm = svm.predict(x_test)
test_acc_svm = accuracy_score(y_test, y_pred_test_svm)
```

In [242…

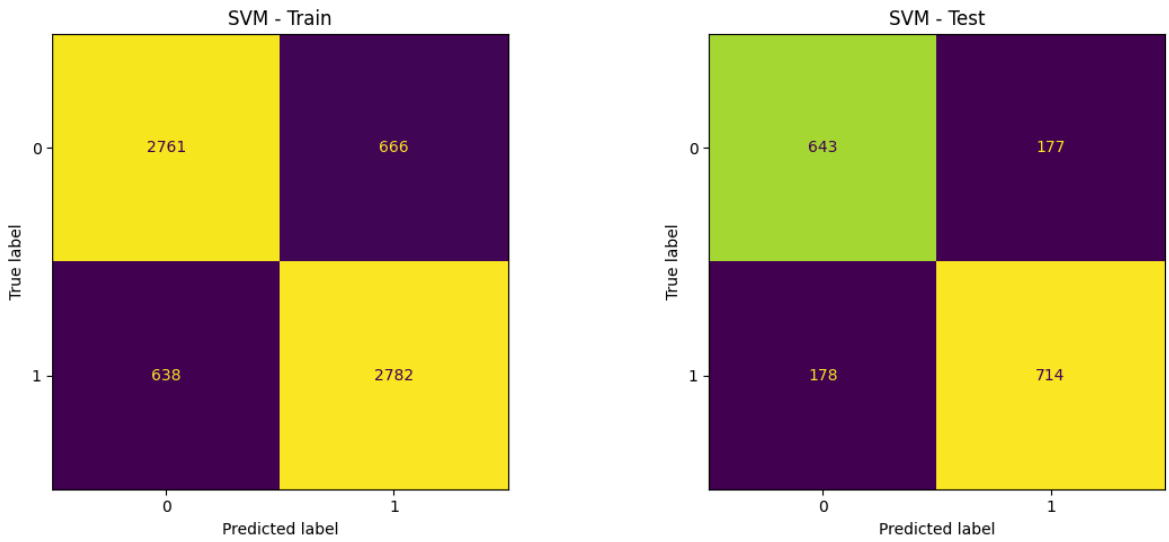
```
print("="*40)
print("BEST SVM – TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_svm*100:.2f}%")
print("="*40)

print("="*40)
print("BEST SVM – TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_svm*100:.2f}%")
print("="*40)
```

```
========================================
BEST SVM – TRAIN SET PERFORMANCE
========================================
Accuracy : 80.96%
========================================
========================================
BEST SVM – TEST SET PERFORMANCE
========================================
Accuracy : 79.26%
========================================
```

## Confusion matrix

In [243…

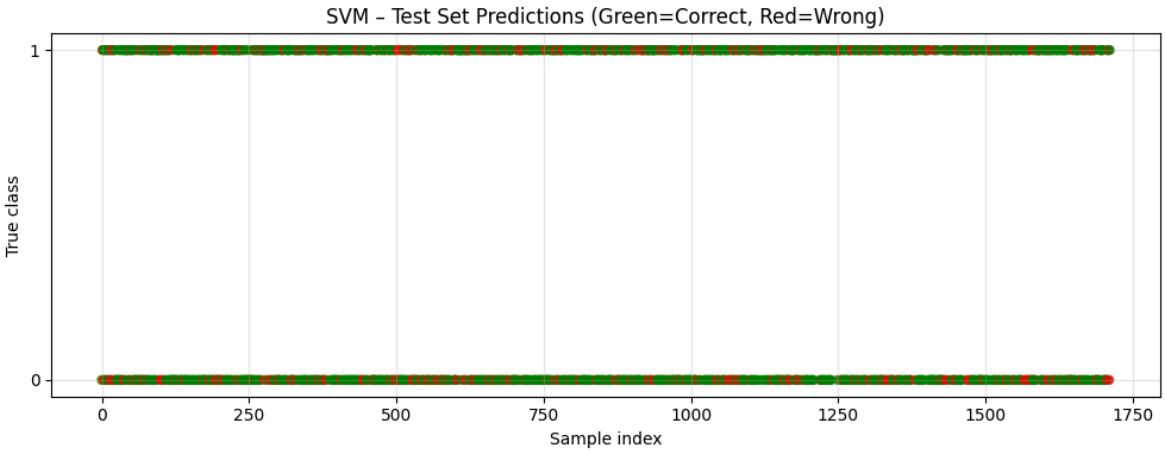
```
plot_confusion_matrices(svm, "SVM", x_train, y_train, x_test, y_test)
```




In [244…

```
correct = (y_pred_test_svm == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
```

```
plt.title("SVM – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"SVM Test Accuracy: {100*test_acc_svm:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
SVM Test Accuracy: 79.26%
Errors: 355 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [245…

```
evaluate_model_metrics(svm, "SVM", x_train, y_train, x_test, y_test)
```

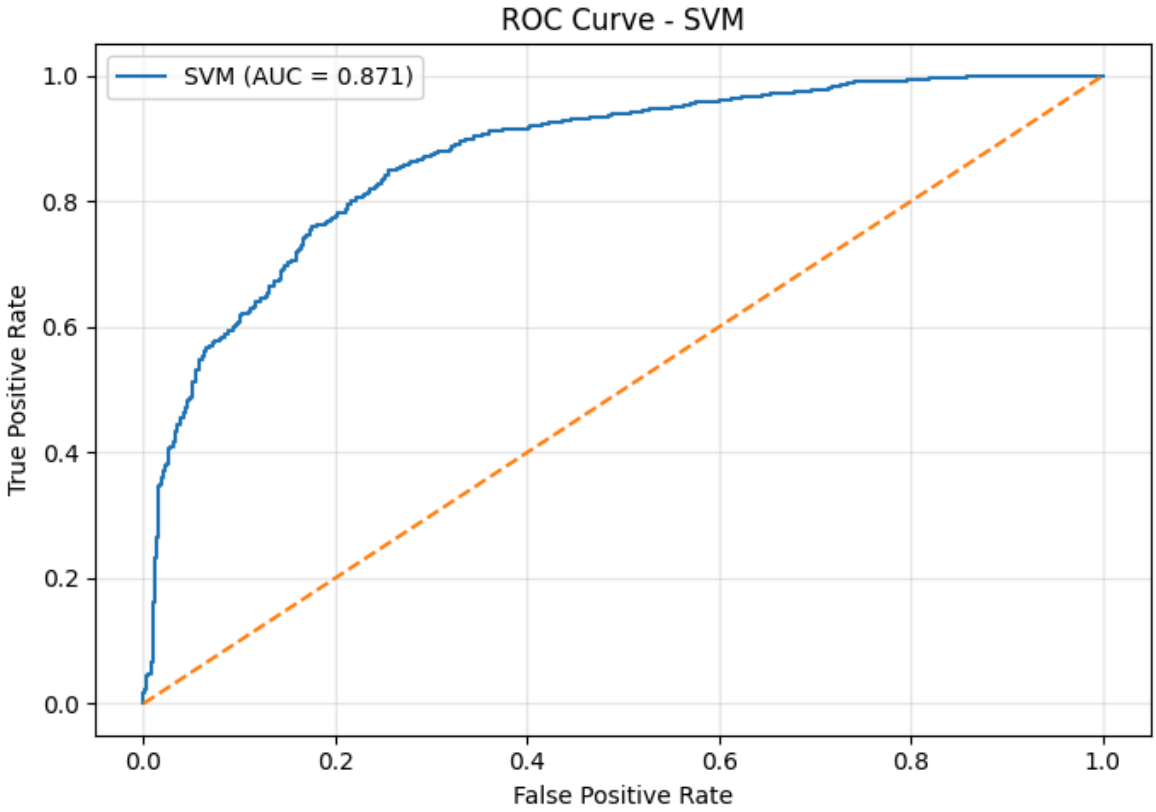
SVM Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 78.32% | 78.41% | 78.37% | 82000.00% |
| **1** | 80.13% | 80.04% | 80.09% | 89200.00% |
| **accuracy** | 79.26% | 79.26% | 79.26% | 79.26% |

## roc curve

In [246…

```
auc_val = plot_roc_curve(svm, "SVM", x_test, y_test)
```




שמירה

In [247…

```
results.append({
    "Model": "SVM",
    "Train Accuracy": train_acc_svm,
    "Test Accuracy": test_acc_svm
})
```

---

# Logistic Regression

## בנייה

In [248…

```
lr = LogisticRegression(
    penalty='l1',
    C=10,
    solver='liblinear',
    max_iter=5000,
    random_state=42
)

lr.fit(x_train, y_train)
```

Out[248…

```
LogisticRegression
LogisticRegression(C=10, max_iter=5000, penalty='l1', random_state=42,
                   solver='liblinear')
```

## דיוק אימון ומבחן

In [249…

```
y_pred_train_lr = lr.predict(x_train)
train_acc_lr = accuracy_score(y_train, y_pred_train_lr)

y_pred_test_lr = lr.predict(x_test)
test_acc_lr = accuracy_score(y_test, y_pred_test_lr)
```

In [250…

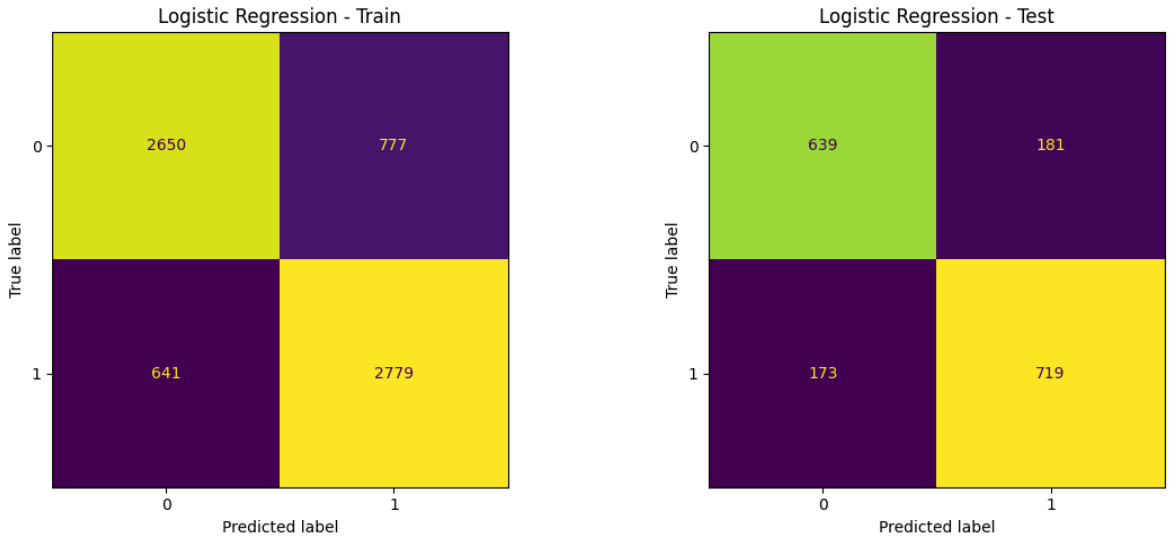
```
print("="*40)
print("BEST LR — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_lr*100:.2f}%")
print("="*40)

print("="*40)
print("BEST LR — TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_lr*100:.2f}%")
print("="*40)
```

```
========================================
BEST LR — TRAIN SET PERFORMANCE
========================================
Accuracy : 79.29%
========================================
========================================
BEST LR — TEST SET PERFORMANCE
========================================
Accuracy : 79.32%
========================================
```

## Confusion matrix

In [251…

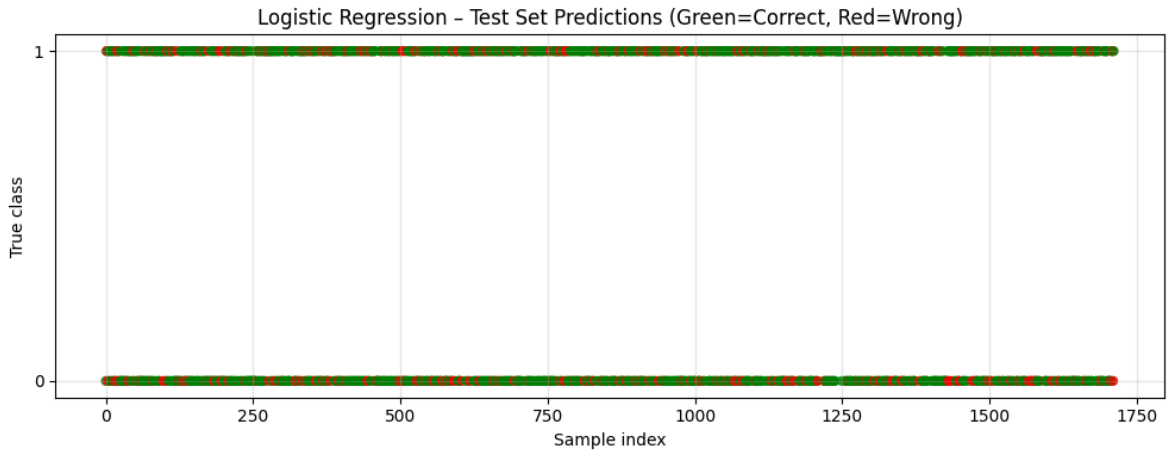
```
plot_confusion_matrices(lr, "Logistic Regression", x_train, y_train, x_test, y_t
```




In [252…

```
correct = (y_pred_test_lr == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
```

```
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("Logistic Regression – Test Set Predictions (Green=Correct, Red=Wrong)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"Logistic Regression Test Accuracy: {100*test_acc_lr:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
Logistic Regression Test Accuracy: 79.32%
Errors: 354 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [253…

```
evaluate_model_metrics(lr, "Logistic Regression", x_train, y_train, x_test, y_te
```

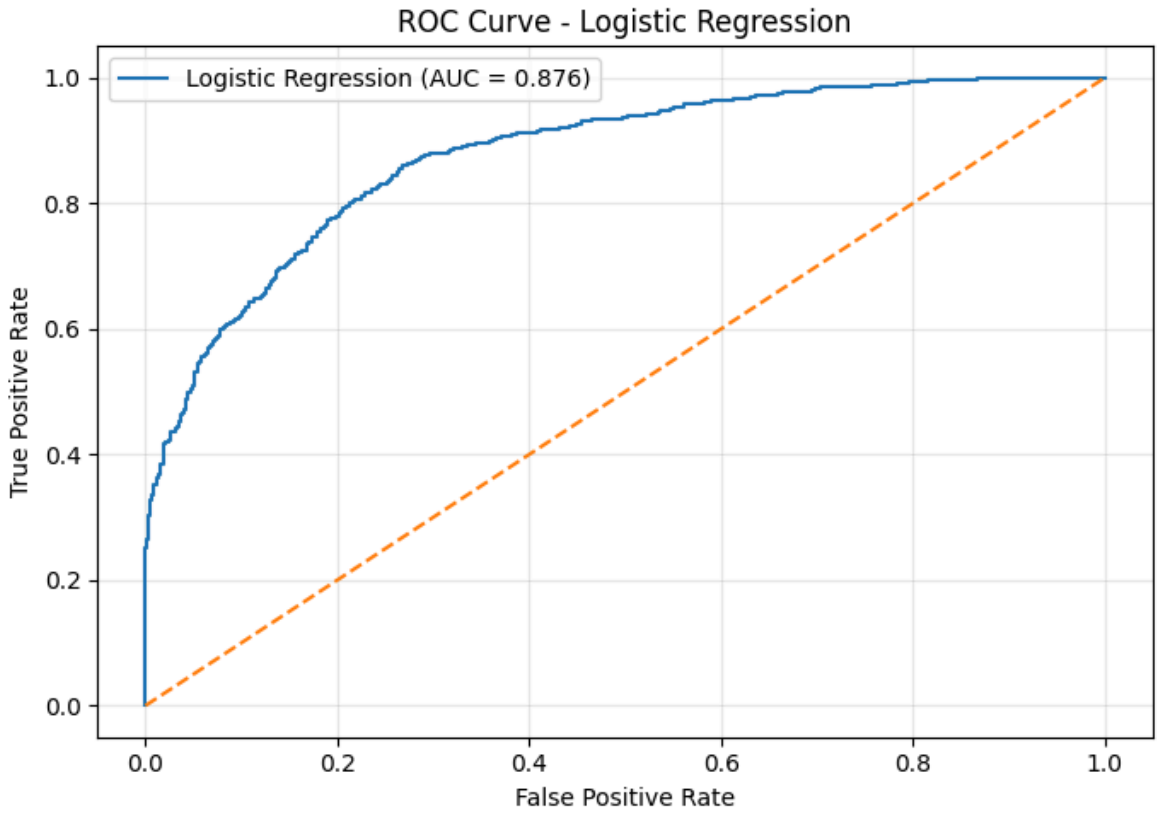
Logistic Regression Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 78.69% | 77.93% | 78.31% | 82000.00% |
| **1** | 79.89% | 80.61% | 80.25% | 89200.00% |
| **accuracy** | 79.32% | 79.32% | 79.32% | 79.32% |

## roc curve

In [254…

```
auc_val = plot_roc_curve(lr, "Logistic Regression", x_test, y_test)
```




שמירה

In [255…

```
results.append({
    "Model": "Logistic Regression",
    "Train Accuracy": train_acc_lr,
    "Test Accuracy": test_acc_lr
})
```

---

# MLP

## בנייה

In [256…

```
mlp = MLPClassifier(
    hidden_layer_sizes=(128, 64),
    alpha=0.01,
    activation="tanh",
    max_iter=5000,
    random_state=42
)
mlp.fit(x_train, y_train)
```

Out[256…

```
MLPClassifier
MLPClassifier(activation='tanh', alpha=0.01, hidden_layer_sizes=(128, 64),
              max_iter=5000, random_state=42)
```

## דיוק אימון ומבחן

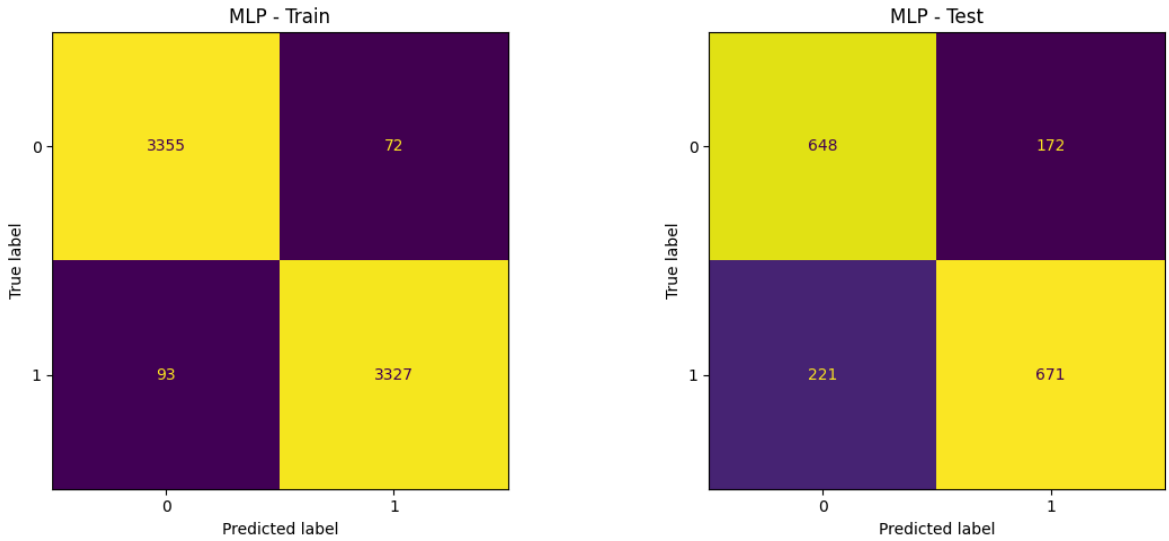
In [257…
```
y_pred_train_mlp = mlp.predict(x_train)
train_acc_mlp = accuracy_score(y_train, y_pred_train_mlp)

y_pred_test_mlp = mlp.predict(x_test)
test_acc_mlp = accuracy_score(y_test, y_pred_test_mlp)
```

In [258…
```
print("="*40)
print("BEST MLP — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_mlp*100:.2f}%")
print("="*40)

print("="*40)
print("BEST MLP — TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_mlp*100:.2f}%")
print("="*40)
```

```
========================================
BEST MLP — TRAIN SET PERFORMANCE
========================================
Accuracy : 97.59%
========================================
========================================
BEST MLP — TEST SET PERFORMANCE
========================================
Accuracy : 77.04%
========================================
```

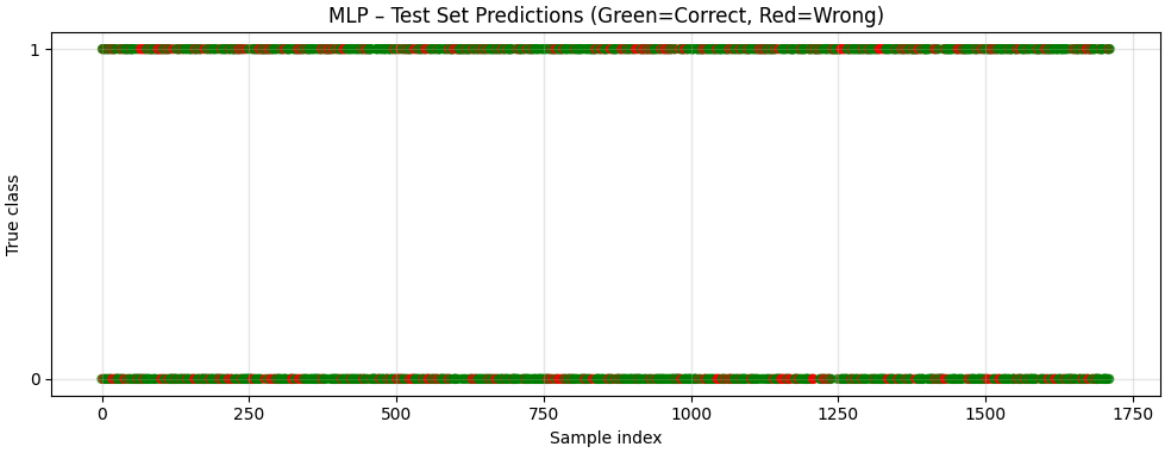
## Confusion matrix

In [259…
```
plot_confusion_matrices(mlp, "MLP", x_train, y_train, x_test, y_test)
```




In [260…
```
correct = (y_pred_test_mlp == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
```

```
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("MLP – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"MLP Test Accuracy: {100*test_acc_mlp:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
MLP Test Accuracy: 77.04%
Errors: 393 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [261…

```
evaluate_model_metrics(mlp, "MLP", x_train, y_train, x_test, y_test)
```

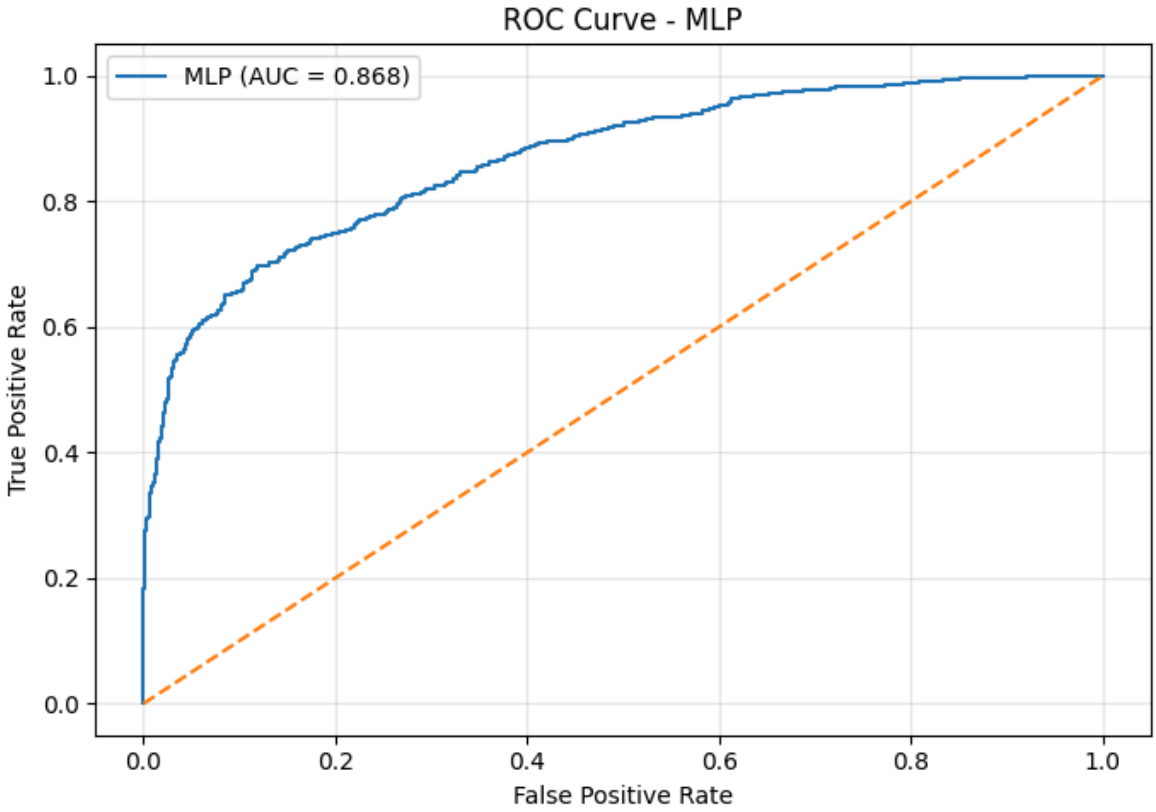
MLP Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 74.57% | 79.02% | 76.73% | 82000.00% |
| **1** | 79.60% | 75.22% | 77.35% | 89200.00% |
| **accuracy** | 77.04% | 77.04% | 77.04% | 77.04% |

## roc curve

In [262…

```
auc_val = plot_roc_curve(mlp, "MLP", x_test, y_test)
```




שמירה

In [263…

```
results.append({
    "Model": "MLP",
    "Train Accuracy": train_acc_mlp,
    "Test Accuracy": test_acc_mlp
})
```

---

# xgboost

## בנייה

In [264…

```
best_xgb_params = {
    "max_depth": 5,
    "n_estimators": 150,
    "learning_rate": 0.1
}
xgb = XGBClassifier(**best_xgb_params, random_state=42)
xgb.fit(x_train, y_train)
```

Out[264…

XGBClassifier

```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=None, device=None, early_stopping_roun
ds=None,
              enable_categorical=False, eval_metric=None, feature_typ
es=None,
              feature_weights=None, gamma=None, grow_policy=None,
              importance_type=None, interaction_constraints=None,
              learning_rate=0.1, max_bin=None, max_cat_threshold=Non
```

## דיוק אימון ומבחן

In [265…

```
y_pred_train_xgb = xgb.predict(x_train)
train_acc_xgb = accuracy_score(y_train, y_pred_train_xgb)

y_pred_test_xgb = xgb.predict(x_test)
test_acc_xgb = accuracy_score(y_test, y_pred_test_xgb)
```

In [266…

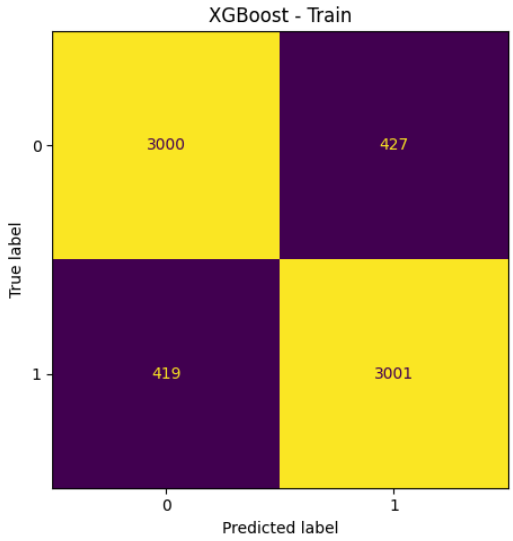
```
print("="*40)
print("BEST XGB — TRAIN SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {train_acc_xgb*100:.2f}%")
print("="*40)

print("="*40)
print("BEST XGB — TEST SET PERFORMANCE")
print("="*40)
print(f"Accuracy : {test_acc_xgb*100:.2f}%")
print("="*40)
```

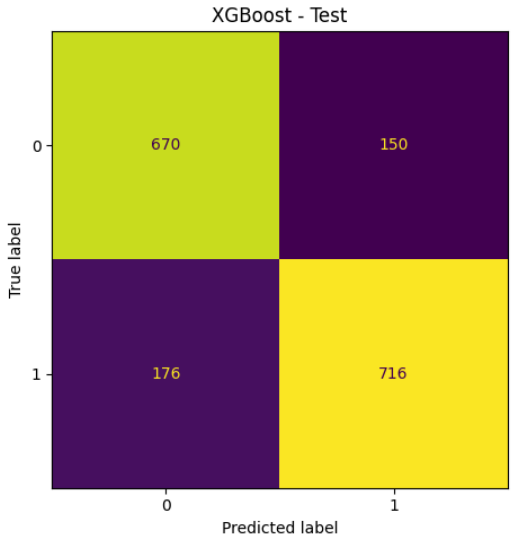
```
========================================
BEST XGB — TRAIN SET PERFORMANCE
========================================
Accuracy : 87.64%
========================================
========================================
BEST XGB — TEST SET PERFORMANCE
========================================
Accuracy : 80.96%
========================================
```

## Confusion matrix

In [267…

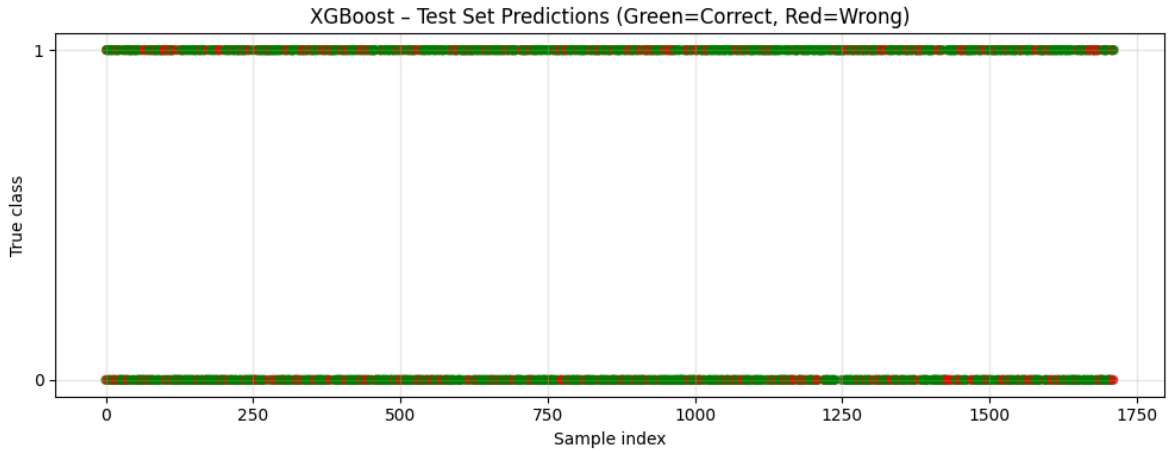
```
plot_confusion_matrices(xgb, "XGBoost", x_train, y_train, x_test, y_test)
```




In [268…

```
correct = (y_pred_test_xgb == y_test)

# צבעים: ירוק = נכון, אדום = שגוי
colors = np.where(correct, "green", "red")

plt.figure(figsize=(10,4))
plt.scatter(range(len(y_test)), y_test, c=colors, s=25, alpha=0.7)

plt.yticks([0,1])
plt.xlabel("Sample index")
plt.ylabel("True class")
plt.title("XGBoost – Test Set Predictions (Green=Correct, Red=Wrong)")
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()

print(f"XGBoost Test Accuracy: {100*test_acc_xgb:.2f}%")
print(f"Errors: {np.sum(~correct)} / {len(correct)}")
```




```
XGBoost Test Accuracy: 80.96%
Errors: 326 / 1712
```

## accuracy_score, precision_score, recall_score, f1_score, roc_auc_score

In [269…

```
evaluate_model_metrics(xgb, "XGBoost", x_train, y_train, x_test, y_test)
```

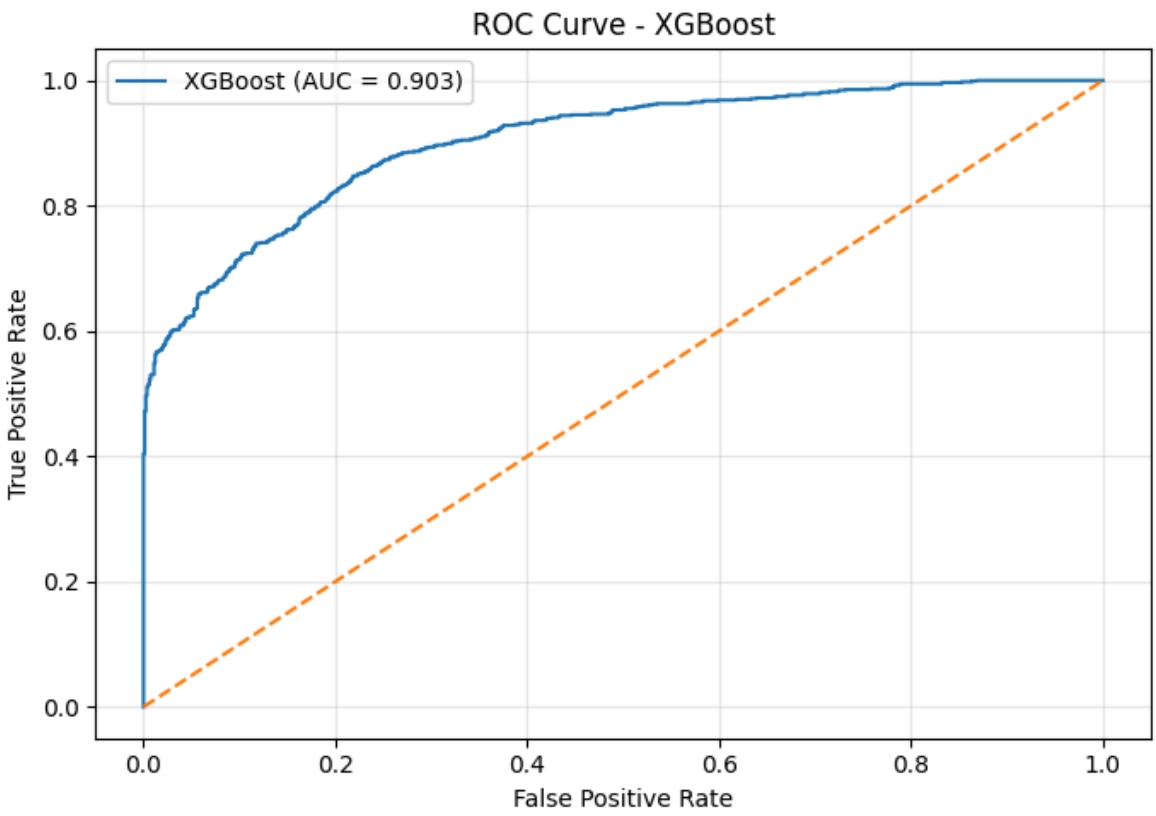
XGBoost Classification Report

| | precision | recall | f1-score | support |
|---|---|---|---|---|
| **0** | 79.20% | 81.71% | 80.43% | 82000.00% |
| **1** | 82.68% | 80.27% | 81.46% | 89200.00% |
| **accuracy** | 80.96% | 80.96% | 80.96% | 80.96% |

### roc curve

In [275…

```
auc_val = plot_roc_curve(xgb, "XGBoost", x_test, y_test)
```




שמירה

In [271…

```
results.append({
    "Model": "XGBoost",
    "Train Accuracy": train_acc_xgb,
    "Test Accuracy": test_acc_xgb
})
```

---

## השוואה סופית

In [272…

```
results_df = pd.DataFrame(results).copy()

sort_col = "Test Accuracy" if "Test Accuracy" in results_df.columns else results
results_df = results_df.sort_values(sort_col, ascending=False).reset_index(drop=

display(
    results_df.style
```

```
        .hide(axis="index")
        .format("{:.2%}", subset=[c for c in results_df.columns if c != "Model"]
        .set_caption("Model Comparison (Train/Test Accuracy)")
)
```

Model Comparison (Train/Test Accuracy)

| Model | Train Accuracy | Test Accuracy |
|---|---|---|
| XGBoost | 87.64% | 80.96% |
| Random Forest | 87.62% | 80.02% |
| Logistic Regression | 79.29% | 79.32% |
| SVM | 80.96% | 79.26% |
| Decision Tree | 79.68% | 77.80% |
| KNN | 79.16% | 77.63% |
| MLP | 97.59% | 77.04% |
| LDA | 76.92% | 76.64% |
| QDA | 76.57% | 76.52% |
| GNB | 72.08% | 72.49% |

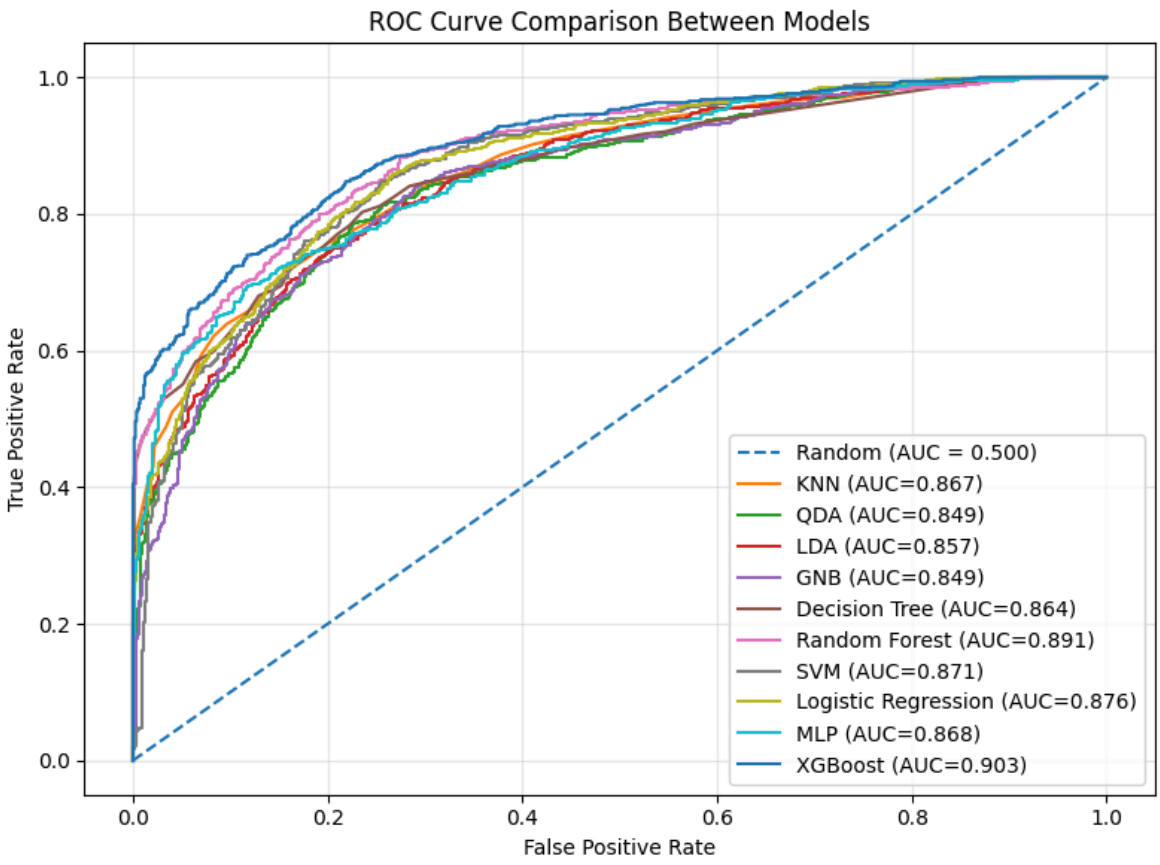
In [273…

```
x = np.arange(len(results_df))
width = 0.35

plt.figure(figsize=(12,5))
plt.bar(x - width/2, results_df["Train Accuracy"], width, label="Train")
plt.bar(x + width/2, results_df["Test Accuracy"],  width, label="Test")

plt.xticks(x, results_df["Model"], rotation=30, ha="right")
plt.ylim(0, 1)
plt.ylabel("Accuracy")
plt.title("Train vs Test Accuracy by Model")
plt.grid(alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()
```

השוואת גרפי roc

In [280…

```
plt.figure(figsize=(8,6))

# baseline
plt.plot([0, 1], [0, 1], "--", label="Random (AUC = 0.500)")

models = {
    "KNN": knn,
    "QDA": qda,
    "LDA": lda,
    "GNB": gnb,
    "Decision Tree": dt,
    "Random Forest": rf,
    "SVM": svm,
    "Logistic Regression": lr,
    "MLP": mlp,
    "XGBoost": xgb
}

for r in results:
    name = r["Model"]
    model = models[name]

    if hasattr(model, "predict_proba"):
        y_score = model.predict_proba(x_test)[:, 1]
    elif hasattr(model, "decision_function"):
        y_score = model.decision_function(x_test)
    else:
        continue

    fpr, tpr, _ = roc_curve(y_test, y_score)
    roc_auc = auc(fpr, tpr)

    plt.plot(fpr, tpr, label=f"{name} (AUC={roc_auc:.3f})")

plt.title("ROC Curve Comparison Between Models")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.grid(alpha=0.3)
plt.legend()
plt.tight_layout()
plt.show()
```



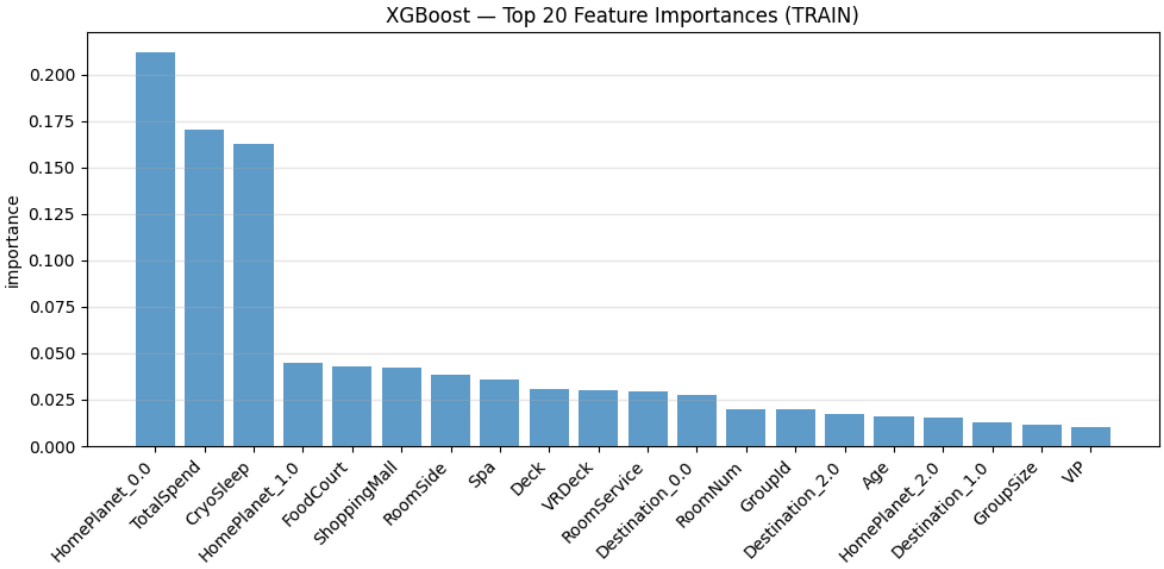

---

הצגת חשיבות של כל תכונה

In [283…

```
feature_names = x_copy.columns.tolist()
importances = xgb.feature_importances_

fi_df = pd.DataFrame({
    "feature": feature_names,
    "importance": importances
}).sort_values("importance", ascending=False)

display(fi_df.head(20))

x = np.arange(len(fi_df.head(20)))
plt.figure(figsize=(10,5))
plt.bar(x, fi_df.head(20)["importance"], alpha=0.7)
plt.xticks(x, fi_df.head(20)["feature"], rotation=45, ha="right")
plt.title("XGBoost — Top 20 Feature Importances (TRAIN)")
plt.ylabel("importance")
plt.grid(axis="y", alpha=0.3)
plt.tight_layout()
plt.show()
```

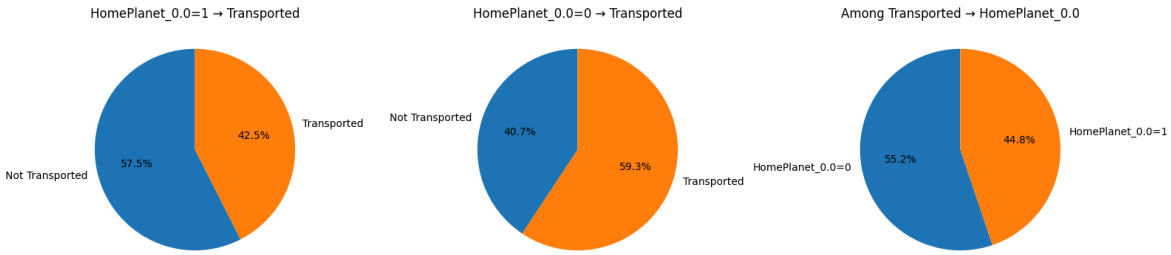
| | feature | importance |
|---|---|---|
| **15** | HomePlanet_0.0 | 0.212128 |
| **8** | TotalSpend | 0.170108 |
| **0** | CryoSleep | 0.162630 |
| **16** | HomePlanet_1.0 | 0.045105 |
| **4** | FoodCourt | 0.042841 |
| **5** | ShoppingMall | 0.042487 |
| **13** | RoomSide | 0.038339 |
| **6** | Spa | 0.035864 |
| **11** | Deck | 0.030750 |
| **7** | VRDeck | 0.029991 |
| **3** | RoomService | 0.029228 |
| **18** | Destination_0.0 | 0.027465 |
| **12** | RoomNum | 0.020037 |
| **9** | GroupId | 0.019578 |
| **20** | Destination_2.0 | 0.017378 |
| **1** | Age | 0.016014 |
| **17** | HomePlanet_2.0 | 0.015218 |
| **19** | Destination_1.0 | 0.013124 |
| **14** | GroupSize | 0.011252 |
| **2** | VIP | 0.010363 |




HomePlanet_0

In [288…

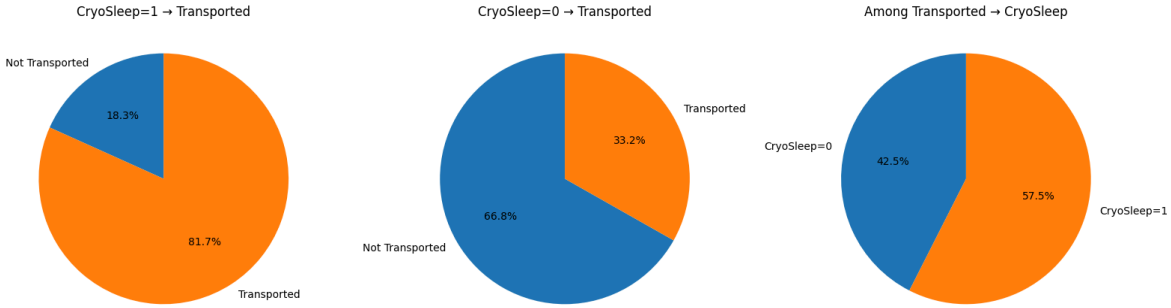
```
plot_feature_transport_pies(spaceship_data, "HomePlanet_0.0")
```




## CryoSleep

In [289…

```
plot_feature_transport_pies(spaceship_data, "CryoSleep")
```

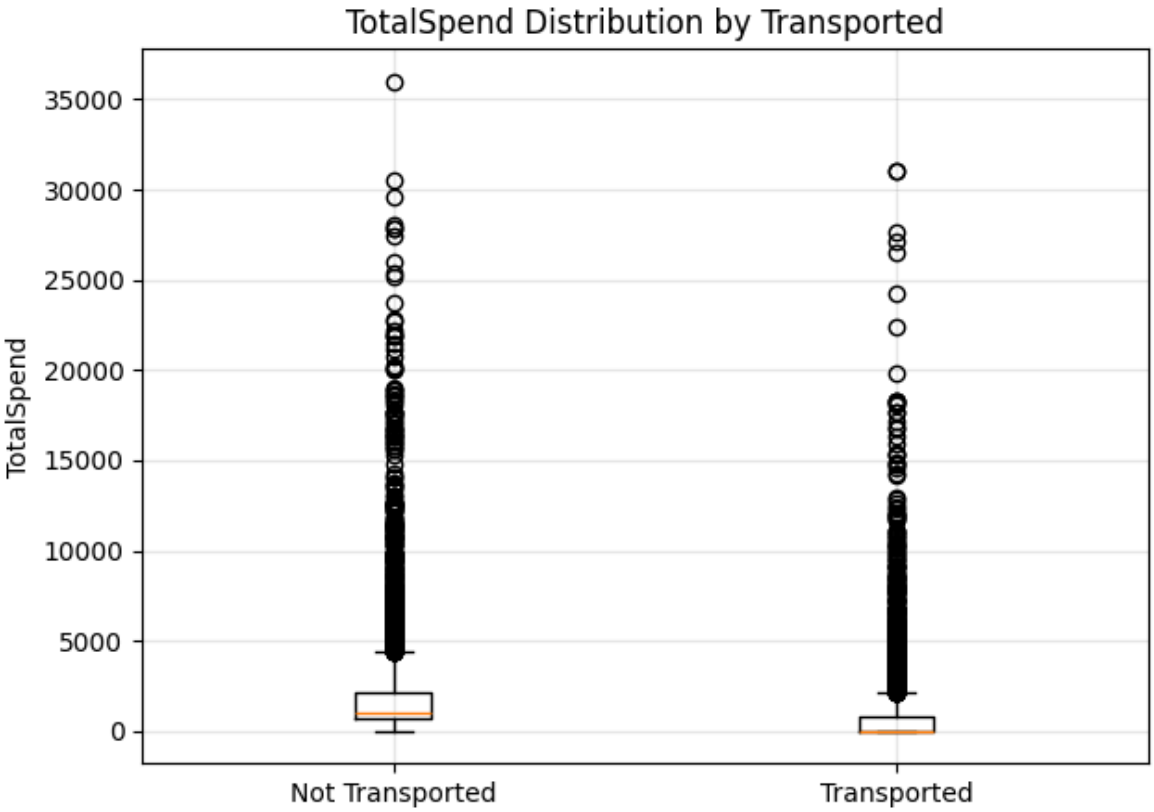



## TotalSpend

In [290…

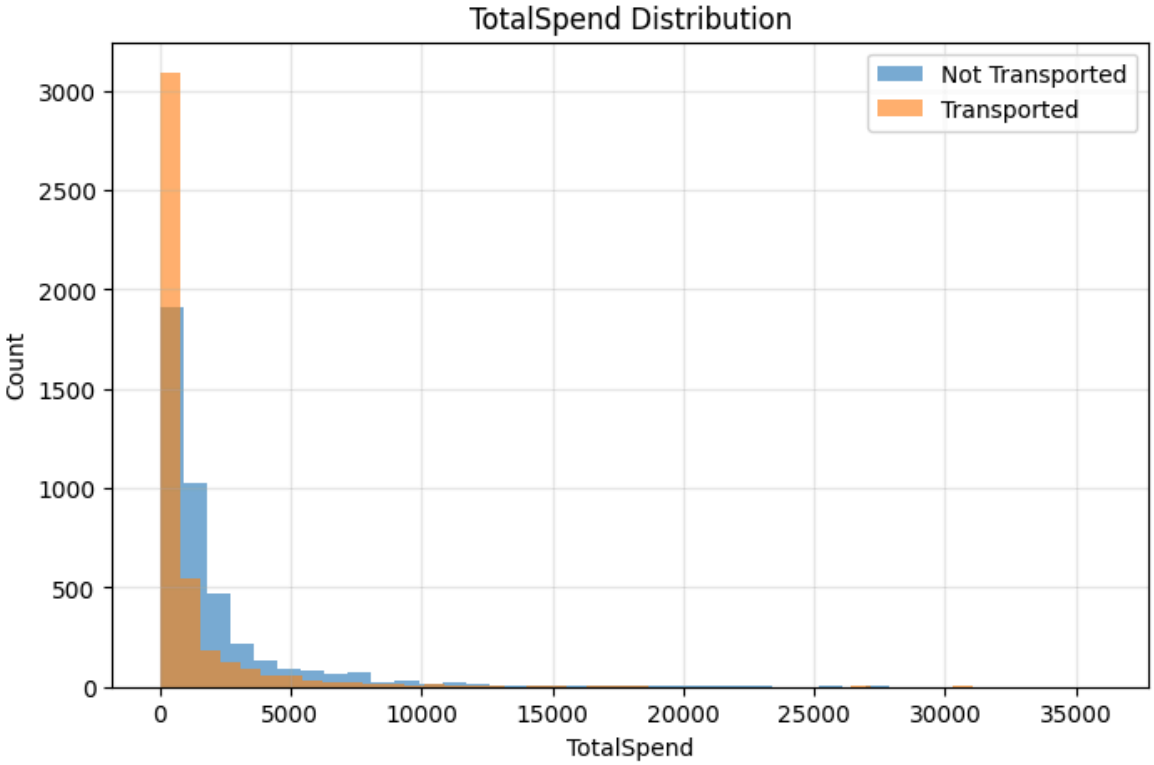
```
plt.figure(figsize=(7,5))

plt.boxplot(
    [
        spaceship_data[spaceship_data["Transported"] == 0]["TotalSpend"].dropna(
        spaceship_data[spaceship_data["Transported"] == 1]["TotalSpend"].dropna(
    ],
    labels=["Not Transported", "Transported"]
)

plt.ylabel("TotalSpend")
plt.title("TotalSpend Distribution by Transported")
plt.grid(alpha=0.3)
plt.show()
```

```
C:\Users\Liad M\AppData\Local\Temp\ipykernel_43036\1771678526.py:3: MatplotlibDep
recationWarning: The 'labels' parameter of boxplot() has been renamed 'tick_label
s' since Matplotlib 3.9; support for the old name will be dropped in 3.11.
  plt.boxplot(
```




In [291…

```
plt.figure(figsize=(8,5))

plt.hist(
    spaceship_data[spaceship_data["Transported"]==0]["TotalSpend"],
    bins=40,
    alpha=0.6,
    label="Not Transported"
)

plt.hist(
    spaceship_data[spaceship_data["Transported"]==1]["TotalSpend"],
    bins=40,
    alpha=0.6,
    label="Transported"
)

plt.xlabel("TotalSpend")
plt.ylabel("Count")
plt.title("TotalSpend Distribution")
plt.legend()
plt.grid(alpha=0.3)
plt.show()
```


TotalSpend Distribution
Not Transported
Transported
Count
3000
2500
2000
1500
1000
500
0
0
5000
10000
15000
20000
25000
30000
35000
TotalSpend